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Bachelor Thesis

zur Erlangung des Grades eines **Bachelor of Science**

Micro-Place: School Density, Student Flows, and Street Crime in Buenos Aires

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Introduction

Every school day in Buenos Aires, roughly 760 thousand students move through the city's street grid in two waves—at entry and at dismissal—generating a predictable, hourly shock to the spatial distribution of potential crime targets. This paper asks whether crime can be predicted by these shocks, and whether it can be suppressed by directed police deployment. The question sits at the intersection of two literatures that rarely speak to each other: the long-run, human-capital channel through which education reduces crime (Lochner and Moretti, 2004; Machin et al., 2011), and the micro-geographic criminology of place, which asks not whether schools reduce crime over a lifetime but whether they concentrate it over a lunch hour.

Three mechanisms operate simultaneously when students move through an urban corridor, and they pull in opposite directions. The **Incapacitation Effect**: during instructional hours, schools physically remove a large share of the youth population—the demographic most overrepresented in street crime offending—from the surrounding street environment, reducing the pool of potential offenders (Anderson, 2014). The **Opportunity Effect**: during entry and dismissal, the mass movement of students creates a predictable concentration of suitable targets in corridors that are too congested for guardianship to be effective (Brantingham and Brantingham, 1995). The **Guardianship Effect**: accompanying adults during school time constitute capable guardianship along school corridors, a protective layer that collapses at dismissal when a share of students are released without organized adult supervision (Cohen and Felson, 1979). The net effect on crime is an empirical question; its sign and magnitude are expected to vary across school types and the city's socioeconomic gradient (Kelly, 2000).

Buenos Aires offers a particularly suitable setting to identify these mechanisms due to its high-density urban morphology. The city's regular orthogonal street grid—*manzanas* of approximately 100 × 100 metres—allows spatial gradients at a granular pedestrian scale. This enables the capture of micro-spatial variation within 100 to 300 metres of a school– the critical distance for foot-traffic interactions. The *Mapa del Delito*, the city's open crime data platform, provides timestamped incident records at street-segment resolution. And the *Senderos Escolares* programme, introduced by the Buenos Aires city government in 2017, deployed specialized policing units along georeferenced school corridors during transition hours —providing a quasi-experimental policy lever to test whether institutionalized guardianship can break the crime-density elasticity documented in the baseline analysis.

To identify the three mechanisms, I construct a panel of 907 grid cells of 500 × 500 metres covering the entire CABA territory, matched to 1,778 geolocated schools with information on management type (public vs. private) and educational level (primary, secondary, and technical-vocational). I exploit within-day variation across four school transition windows—morning entry, midday dismissal and afternoon dismissal—to estimate the spatial gradient of crime around schools at each institutional transition point. Given the preponderance of zero-count observations inherent to fine-grained spatial crime data, I estimate a Poisson fixed-effects model with grid and day fixed effects. Robustness checks employ a finer 300 × 300 metre grid and alternative measures of school exposure based on distance bins and continuous buffer coverage.

My main findings are threefold. First, the spatial gradient of crime around schools is asymmetric across transition windows: at entry hours, grid cells with the centroid within 100 metres of a school record robbery counts 17–21 percent lower than cells whose centroids are 300m or more from the nearest schools, consistent with the guardianship mechanism; at midday dismissal, the gradient reverses, with proximate cells recording counts approximately 11–14 percent *higher*. Gradients don't seem to decay monotonically with distance and are robust to alternative grid resolutions and standard error clustering strategies. Second, aggregate school density exerts at most weak effects on crime: the interaction $N_g^{\text{schools}} \times$ `is_school_time` is statistically indistinguishable from zero across almost all specifications, and only isolated coefficients reach marginal significance when disaggregating by school sector and conditioning on neighborhood poverty. This pattern suggests that what drives the school-hour crime gradient is not the accumulation of schools within a grid but the proximity to the nearest institutional anchor—consistent with a point-source model of pedestrian flow concentration rather than an area-level density effect. Third, a dose-based Sun-Abraham event study of the *Senderos Escolares* programme—requiring a minimum of 200 metres of operational route presence within each grid cell—yields statistically significant post-treatment crime reductions concentrated in the first year of implementation, with the effect attributable to prevention agents rather than the additional police layer.

This paper contributes to three strands of literature. First, it extends the micro-place criminology of Weisburd et al. (2012) to a Latin American urban context, demonstrating that the concentration of crime responds to institutional activity patterns at sub-neighbourhood scales in a city whose street morphology facilitates identification at pedestrian resolution. Second, it contributes to the literature on the immediate spatial externalities of schools (Lochner and Moretti, 2004), providing a design that separately identifies the guardianship and opportunity channels across transition windows rather than treating school proximity as a static characteristic of place. Third, it provides causal evidence on a place-based urban safety intervention in a developing-country context, complementing recent work on targeted policing in Latin American cities (Di Tella and Schargrodsky, 2004; Blattman et al., 2021).

The remainder of this paper is organised as follows. Section 1 describes the data and identification setting. Section 2 presents the empirical strategy. Section 3 reports the main results. Section 4 discusses limitations, and Section 5 concludes.

1 Data

1.1 Spatial Unit of Analysis

The unit of observation is a 500×500 meter grid cell constructed by partitioning the administrative boundary of the Ciudad Autónoma de Buenos Aires (CABA) into a regular lattice. After clipping to the city boundary, the grid yields $N = 907$ cells. This spatial resolution reflects the pedestrian catchment area of a school anchor point, broadly consistent with the behavioral distances documented in Bernasco and Block (2011), while remaining coarse enough to ensure sufficient crime counts in most cells. Robustness checks repeat all specifications on 300-meter grids (2,436 cells). All geographic objects are projected to POSGAR 2007 / Argentina Zone 4 (EPSG:32721), a metric coordinate system appropriate for Euclidean distance calculations within the Buenos Aires metropolitan area. Crime counts are disaggregated by four within-day time slots (*franjas horarias*), yielding an estimation sample of $907 \times 211 \times 4 = 765{,}508$ grid $\times$ day $\times$ slot observations for 2016; the event-study extends to 2019, excluding lockdown months. Roughly 95 percent of observations record zero crimes, motivating the Poisson FE estimator (Osgood, 2000).

1.2 Data Sources and Variable Construction

All datasets are obtained from the Buenos Aires City open data platform (*Buenos Aires Data*) and are publicly available.

Outcome: Street Crime

The dependent variable is constructed from the *Mapa del Delito*, which records all crimes reported to the Buenos Aires City Police. Each record contains geocoordinates, date, hour-of-day slot (*franja horaria*), and offense classification. The primary outcome aggregates street robberies (*Robo*) and non-contact thefts (*Hurto*), as these offense types are the most sensitive to pedestrian density and target availability (Bernasco and Block, 2011). Heterogeneous effects by offense type are examined in supplementary specifications.

School Density and Proximity

Schools are obtained from the *Establecimientos Educativos* dataset, which lists all operating schools in CABA with addresses, educational level (*nivel modal*), and sector (public/*Estatal* vs. private/*Privado*). Coordinates were geocoded from addresses; approximately 97 percent of records achieved a valid match. The remaining 3 percent of unmatched records—concentrated in ambiguous addresses and schools located within larger institutional complexes—are dropped from the estimation sample. Given the dense spatial coverage of CABA's educational infrastructure (99 percent of grid cells lie within 500 m of a matched school), the exclusion of 3 percent of schools is unlikely to produce systematic attenuation in the exposure measures. Schools are classified as primary (*primario*) or secondary (*secundario*), with sector recorded as public (*Estatal*) or private (*Privado*).

The main school exposure variable, *n_schools*, counts the number of schools whose centroid falls within each grid cell. For non-parametric analysis of proximity effects, I compute the Euclidean distance from each grid centroid to the nearest school and assign it to one of four bins: 0–100 m, 100–200 m, 200–300 m, and above 300 m (reference category), following the buffer radio used in Bernasco and Block (2011). Distances are measured from grid centroids rather than from cell boundaries: the centroid provides a single representative point for each spatial unit, minimising sensitivity to the arbitrary orientation of the grid relative to the school network. For square cells of 500×500 m, the maximum within-cell deviation from the centroid is approximately 354 m (half-diagonal), smaller than the 500 m buffer used in the robustness grid and comparable to the width of the outermost distance bin. Sensitivity to boundary effects at cells intersecting the administrative perimeter of CABA is examined in the border-grid robustness check (Section 2).

Treatment: ***Senderos Escolares***

The *Senderos Escolares* program assigns pedestrian safety agents (*Agentes de Prevención*) to designated school routes throughout CABA. Routes are available as `MULTILINESTRING` geometries. The treatment date corresponds to the first route crossing a cell, yielding two main adoption cohorts: Phase 1 (2017) and Phase 2 (2021). Routes formally introduced in 2020 are reclassified to 2021, since the national lockdown decree (DNU 297/2020, enacted March 20, 2020) suspended program operations throughout the ASPO period. The vintage of the routes dataset is 2024; the resulting limitation — that the pre-adoption control group may be attenuated if later cohorts are already treated in earlier years — is discussed in Section 2.

Controls

Socioeconomic conditions. The poverty rate is defined as the share of households with at least one Unmet Basic Need (*NBI*), obtained from the 2010 Argentine Population Census at the *radio censal* level. Since census radios do not align with the 500-meter grid, the variable is interpolated to the grid using areal weighting: each radio's value is distributed to overlapping cells in proportion to the share of the radio's area falling within the cell. Population counts serve as an exposure offset in count models.

Land use. Parcel-level land use from the *Relevamiento de Uso del Suelo* dataset is aggregated to two grid-level proportions: *pct_commercial* (share of parcels classified as commercial retail) and *pct_gastronomy* (share classified as gastronomy, bars, or entertainment). These variables proxy for ambient economic activity following the routine activity framework (Cohen and Felson, 1979).

Infrastructure density. Counts of police stations (*comisarías*) and bus stops (*paradas de colectivo*) per grid cell are included as placebo controls. Both are time-invariant within 2016 and therefore absorbed by grid fixed effects; they are relevant only as interactions and placebo checks, as described in Section 2.

1.3 Summary Statistics

Table 1 reports summary statistics for the main analysis variables. Crime is highly concentrated spatially: considering all offense types recorded in the *Mapa del Delito*, the mean grid cell records approximately 1,338 incidents over 2016, but the median is 994 and the distribution is strongly right-skewed (max: 11,579), consistent with the hot-spots literature (Sherman et al., 1989). Restricting to the analysis outcome (robberies and non-contact thefts), the mean grid cell records 49 incidents over 2016 (median: 31, max: 648), reflecting the concentration of street crime in a small number of high-activity cells. Approximately 66 percent of grid cells contain at least one school, with an average of 2.08 schools per cell and a maximum of 19. The median distance to the nearest school is 217 meters, and 99 percent of cells lie within 500 meters of a school, reflecting the dense educational infrastructure of the city. The poverty rate averages 6.2 percent but is highly skewed, with a 90th-percentile value of 18.2 percent; grids with missing census data are concentrated in non-residential areas (Reserva Ecológica, Puerto Madero), indicating structurally missing rather than random absence.

The hourly crime distribution exhibits a pronounced evening concentration and a secondary inflection at 12:00–13:00 coinciding with primary school dismissal—the empirical regularity that motivates the within-day identification strategy (Figure 1).

Table 1: Summary Statistics

Variable	Mean	Std. Dev.	Median	Max
Crime (2016)				
Annual crimes per grid cell (all offenses)	1,338	–	994	11,579
Annual robberies + thefts per grid cell	49	68.3	31	648
Daily crimes (city total)	348.35	83.90	364	437
School exposure				
Schools per grid cell	2.08	–	2	19
Grids with ≥ 1 school (%)	65.85	–	–	–
Distance to nearest school (m)	337	351	217	2,207
Within 500m of a school (%)	99.3	–	–	–
Total schools (CABA)	1,793	–	–	–
Public (*Estatal*)	1,083	–	–	–
Private (*Privado*)	710	–	–	–
Primary level	879	–	–	–
Secondary level	495	–	–	–
Census variables (radio censal level)				
Population	813	293	786	1,093
Poverty rate (NBI)	0.062	0.096	0.019	0.182

Note: $N = 907$ grid cells.

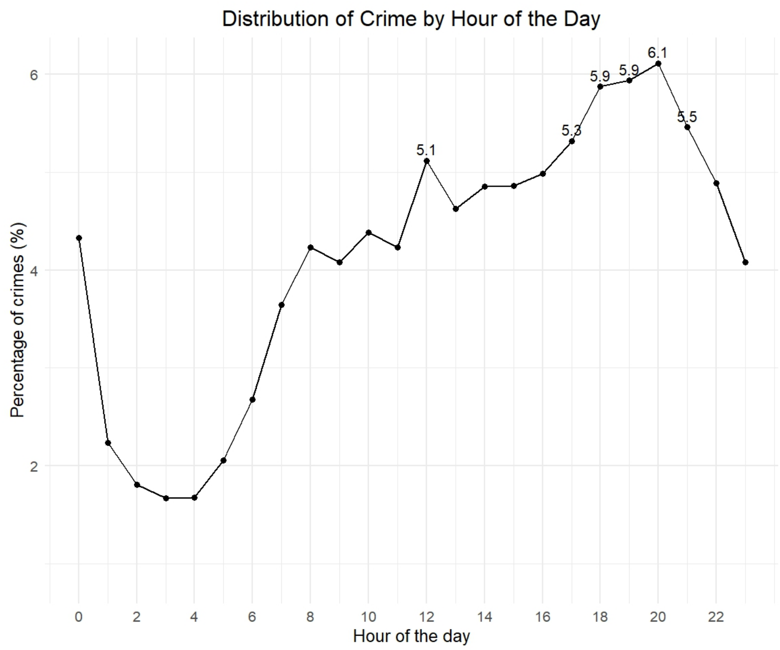

Figure 1: Distribution of criminal reports by hour of the day, CABA 2016. The midday inflection (12:00–13:00) coincides with the intershift dismissal period of CABA's school system. The dominant peak at 18:00–20:00 reflects the broader evening crime cycle. Source: Mapa del Delito, Ministerio de Seguridad CABA.

The pronounced north–south poverty gradient (Figure 2) motivates the heterogeneity analysis in Section 3: the school's role as crime attractor or safe anchor is conditional on the socioeconomic fabric of the surrounding neighbourhood.

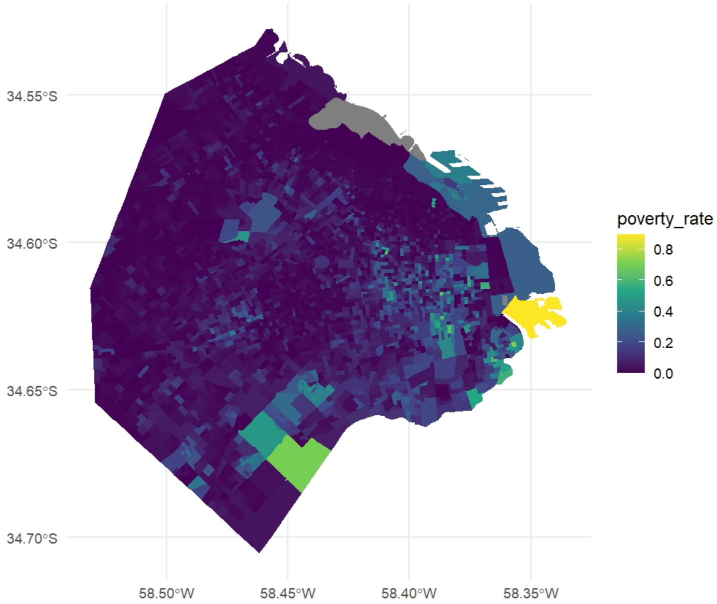

Figure 2: Poverty rate (NBI) by census tract (*radio censal*), CABA 2010. High-poverty concentrations in the southeastern districts and informal settlements along the Riachuelo contrast with the low-poverty northern and central districts. Source: INDEC, Censo Nacional 2010.

Crime concentrates on weekdays with a pronounced Sunday trough (Figure 3), consistent with routine activity theory: institutional schedules generate target-rich environments that collapse when schools and commuting patterns are absent.

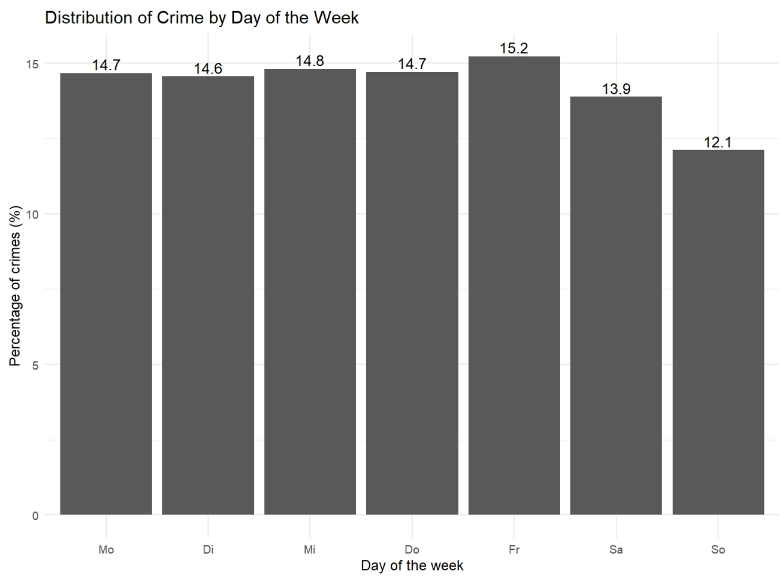

Figure 3: Distribution of criminal reports by day of the week, CABA 2016. Weekday concentration is consistent with routine activity theory. Source: Mapa del Delito, Ministerio de Seguridad CABA.

The January–February trough in monthly crime counts reflects the summer school vacation period, during which student flows cease and the within-day identification variation is absent; the sharp contraction from March 2020 onward reflects the national lockdown

(ASPO, DNU 297/2020), which suspended both economic activity and *Senderos Escolares* operations, motivating the restriction of the event-study panel to 2016–2019 (Figure 4).

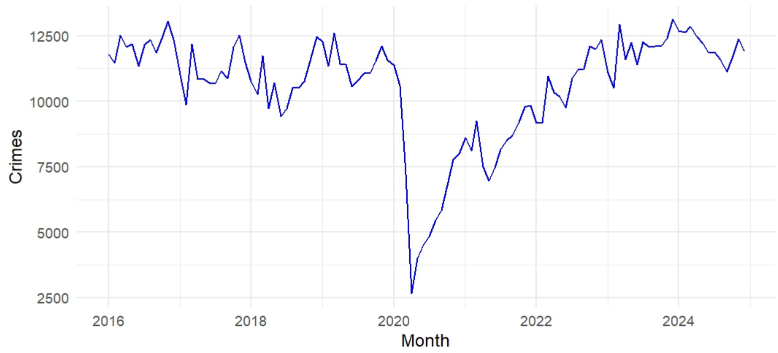

Figure 4: Monthly distribution of criminal reports, CABA 2016. Source: Mapa del Delito, Ministerio de Seguridad CABA.

2 Empirical Strategy

2.1 Identification Framework

The core identification challenge is that schools are not randomly located: denser, more commercially active neighbourhoods tend to cluster both more schools and more crime. A naive cross-sectional regression of crime on school counts would therefore confound the pedestrian-concentration effect with underlying neighbourhood characteristics. I address this through a within-grid, within-day design that exploits the deterministic temporal variation in student presence rather than cross-sectional variation in school location.

The key insight is that school buildings are fixed, but the flow of students, parents, and accompanying pedestrian activity they generate is highly time-structured. Comparing crime rates at the *same* grid cell between hours when students are physically outside schools and hours when they are inside—controlling for the daily crime trend common to all cells—isolates the marginal effect of student pedestrian concentration around schools on criminal opportunity.

2.2 Baseline Specification

The unit of observation is a grid cell g on day d in time slot t. The panel covers all $500\,\text{m} \times 500\,\text{m}$ grid cells of CABA ($G = 907$), across 211 weekdays of 2016 (January and February excluded due to summer school vacation), restricted to the four time slots defined in Section 1. The outcome Y_{gdt} is the count of robberies (or thefts, depending on specification) recorded in cell g, day d, slot t.

I estimate a two-way fixed-effects Poisson (TWFE-Poisson) model via `fepois` from the `fixest` package (Bergé, 2024):

$$\mathbb{E}[Y_{gdt} \mid \mathbf{x}_{gdt}] = \exp\Big(\beta_1 \,\text{SchoolHour}_t + \beta_2 \, N_g^{\text{schools}} \times \text{SchoolHour}_t + \alpha_g + \delta_d\Big) \qquad (1)$$

where $\text{SchoolHour}_t \in \{0, 1\}$ equals one during school entry/exit slots and zero during control slots; N_g^{schools} is the count of schools in cell g; α_g are grid fixed effects absorbing all time-invariant neighbourhood attributes (location, land use, poverty, permanent police presence, etc.); and δ_d are day fixed effects absorbing city-wide daily shocks (weather, events, day-of-week patterns). Standard errors are clustered at the grid level throughout; Appendix A reports Conley spatial standard errors and two-way clustering as robustness checks.

The coefficient of interest is $\hat{\beta}_2$: the within-grid, within-day semi-elasticity of crime to student density at school hours relative to control hours. Because the Poisson log-link is used, $(\exp(\hat{\beta}_2) - 1) \times 100$ gives the percentage change in expected crime counts per additional school in the cell during school hours relative to the same cell at control hours on the same day.

Heterogeneity by school sector (public vs. private) and neighborhood poverty is examined in Section 3 through disaggregated specifications and triple interactions with the poverty rate.

Identifying assumption. Identification of β_2 rests on the assumption that, conditional on grid fixed effects α_g and day fixed effects δ_d, the interaction $N_g^{\text{schools}} \times \text{SchoolHour}_t$ is uncorrelated with unobserved grid-day-hour shocks to crime. Equivalently, no time-varying factor at the sub-daily level systematically co-moves with school transition hours and concentrates differentially in cells with more schools. Grid fixed effects absorb all time-invariant cell-level confounders; day fixed effects absorb city-wide daily shocks. The residual threat is a grid-day-hour shock specific to school-hour windows and correlated with school density—a scenario discussed in Section 4.

Plausibility of within-day exogeneity. The school schedule in CABA is set centrally by the Buenos Aires City Ministry of Education and is uniform across all public schools within each educational level. Entry and dismissal times are fixed by decree at the start of the academic year and do not respond to local crime conditions. This institutional rigidity is the source of identifying variation: the *timing* of student flows is pre-determined with respect to local crime shocks, satisfying the within-day exogeneity condition required for causal interpretation of $\hat{\beta}_2$. Residual threats to this assumption are discussed in Section 4.

Control vs. treatment time slots. The mapping from raw hourly crime records to the binary SchoolHour_t is:

Category	Hours	Rationale
School hours	7h, 12h–13h, 16h	Entry and primary/secondary exit
Control hours	6h, 8h–11h, 14h–15h, 17h	Street activity comparable
Excluded	0h–5h, 18h–23h	Night/evening; different crime ecology

Table 2: Time-slot classification. Excluded hours are dropped from the estimation sample to preserve a clean counterfactual.

The control group is intentionally restricted to morning and early afternoon hours when students are inside school buildings. Including nighttime as control would conflate the school-hours effect with the well-documented night/day crime cycle (Cohen and Felson, 1979). Alternative school-hour window definitions are examined in Appendix A.5 as a robustness check.

2.3 Spatial Gradient Specifications

To test whether the pedestrian-concentration effect decays with distance from the nearest school—a key prediction of crime pattern theory (Bernasco and Block, 2011)—I replace the linear school-count interaction with a distance-bin design. Let $\text{DistBin}_g \in \{[0,100),[100,200),[200,300),300+\}$ metres denote the Euclidean distance from the centroid of cell g to the nearest school, and let $\mathbf{1}[\cdot]$ denote an indicator. The spatial gradient model is:

$$\mathbb{E}[Y_{gdt}] = \exp\left(\sum_{t'\in\mathcal{T}} \gamma_{t'}\mathbf{1}[\text{Franja}_t = t'] + \sum_{t'\neq\text{control}} \sum_{b\neq 300+} \lambda_{t',b}\,\mathbf{1}[\text{Franja}_t = t'] \times \mathbf{1}[\text{DistBin}_g = b] + \alpha_g + \delta_d\right) \tag{2}$$

The reference categories are the control slot and the 300+ metre bin. The coefficients $\hat{\lambda}_{t',b}$ trace out—for each school time slot $t' \in \{\text{start}, \text{exit\_primaria}, \text{exit\_secundaria}\}$—how much crime at a cell within b metres of a school exceeds crime at a cell beyond 300 m, relative to the same comparison at control hours. Monotone decay of $|\hat{\lambda}_{t',b}|$ as b increases is the spatial analogue of the routine activity prediction: guardianship and target density fall with distance.

An extended version of equation (2) adds the analogous distance-bin interactions for the nearest bus stop and nearest police station, isolating the school proximity effect from confounding infrastructure gradients.

2.4 Buffer Count and Exposure Index

As a complementary operationalisation, I construct for each grid cell the count of schools within concentric buffers of radio 100 m, 200 m, and 300 m:

$$N_g^{\text{buf},r} = \#\{s \in \mathcal{S} : \text{dist}(g,s) \leq r\} \tag{3}$$

and an inverse-distance exposure index:

$$\text{Exposure}_g^r = \sum_{s\in\mathcal{S},\,\text{dist}(g,s)\leq r} \frac{1}{\text{dist}(g,s)} \tag{4}$$

Both enter equation (1) interacted with SchoolHour_t or with the full franja indicators. Unlike the distance-bin design—which assigns each grid cell to a single bin based on its nearest school—buffer counts and the exposure index capture the *intensity* of school presence in the neighbourhood of g, allowing for a dose-response interpretation. Buffers use school point locations (geocoded centroids) and are computed in the POSGAR 2007 / Argentina Zona 4 projected CRS (EPSG: 32721) so that all distances are in metres.

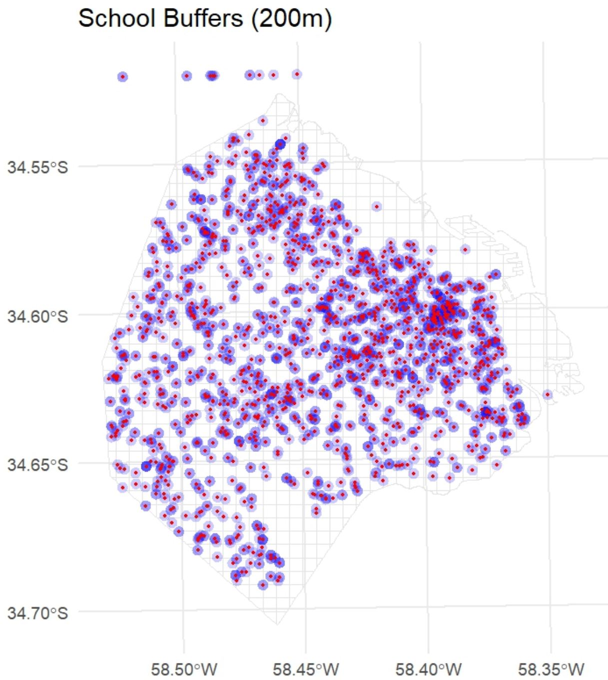

Figure 5: School buffer zones at radio 100 m, 200 m, and 300 m around geocoded school centroids, CABA. Buffers are computed in POSGAR 2007 / Argentina Zona 4 (EPSG: 32721).

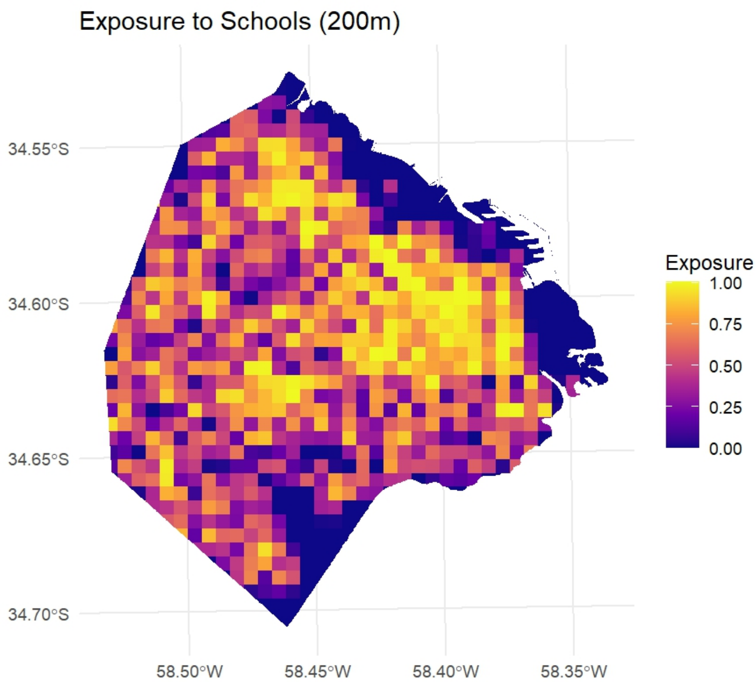

Figure 6: Inverse-distance school exposure index Exposure_g^{300} by grid cell, CABA 2016. Higher values indicate greater density of school presence within 300 m of the cell centroid.

2.5 Robustness

Standard error robustness I report four SE specifications for the main distance-bin and buffer models: (i) cluster by grid cell, (ii) two-way cluster by grid cell and day, (iii) Conley spatial HAC at 0.5 km cutoff, and (iv) Conley at 1 km cutoff (500 m and 300 m for the 300 m grid specifications). Results are qualitatively unchanged across all four.

Grid-size robustness. All main specifications are replicated on a 300 m × 300 m grid partition (1,847 cells), which reduces spatial aggregation bias and allows finer distance bins. Coefficients are comparable in sign and magnitude; significance is maintained.

Border-grid robustness. Grid cells intersecting the administrative perimeter of CABA are geometrically truncated, potentially understating crime counts and inflating buffer-based school exposure if AMBA schools enter the buffer radius. All main specifications are replicated on the subsample of strictly interior cells, with buffer counts recomputed after excluding schools located in border cells. Results are stable; see Appendix A.4.

2.6 Senderos Escolares: Identification Strategy

2.6.1 Programme and Treatment Definition

The *Senderos Escolares* programme designates pedestrian routes linking residential areas to school entrances, accompanied by physical improvements (lighting, sidewalk repairs, signage) and the deployment of *agentes de tránsito* during school entry and exit windows. Our analysis focuses on the **original phase of the programme** ("Fase Original"), operationalised as the 207 routes with the lowest identifier numbers in the administrative dataset (id ≤ 207). Routes with id > 207 correspond to a 2021 expansion and serve as the not-yet-treated control group.[1]

Treatment is assigned at the grid-cell level using the physical length of Sendero network intersecting each cell. A cell g is classified as **treated** if the total metres of Fase Original route lines falling within the cell exceed an operational presence threshold ℓ^*:

$$\text{Treated}_g = \mathbf{1}[L_g > \ell^*], \qquad L_g = \sum_{s \leq 207} \text{length}(s \cap g) \tag{5}$$

where L_g is computed via line intersection in POSGAR 2007 / Argentina Zona 4 (EPSG: 32721). The baseline threshold is $\ell^* = 200$ m, corresponding approximately to two city blocks of Sendero coverage within the cell—the minimum length at which an *Agente de Prevención* stationed on the route generates meaningful visual coverage of the grid. The robustness of results to alternative thresholds ($\ell^* \in \{50, 100, 300, 400\}$ m) is examined in Section 3.4. Never-treated grids ($L_g = 0$ for both phases) are excluded from the estimation sample; the Fase 2 expansion grids serve as not-yet-treated controls throughout the 2016–2019 window.

Threshold selection. The baseline threshold $\ell^* = 200$ m is chosen on operational grounds: a single *Agente de Prevención* stationed at the midpoint of a 200 m segment generates line-of-sight coverage of approximately one full city block in each direction, the

[1] The cutoff id = 207 is a methodological simplification adopted in the absence of official phase documentation. The partition is consistent with the 2017 rollout chronology observable in the data.

minimum spatial unit at which guardianship is credibly continuous. Cells with $L_g < 200\,\text{m}$ contain only a corner segment of the route and are unlikely to experience meaningful enforcement exposure; coding them as treated would introduce classical measurement error that attenuates estimates toward zero.

Pre-treatment parallel trends. Figure 24 plots mean robbery counts per grid cell for Fase 1 and Fase 2 grids across the six pre-treatment bimesters (March–December 2016). Trends are visually parallel, and a joint Wald test on cohort × bimester interactions fails to reject equality of pre-trends, supporting the parallel trends assumption underlying the Sun-Abraham estimator.

2.6.2 Temporal Resolution and Hour Slot Classification

The causal identification exploits within-day variation in enforcement intensity. Ipartition the 14 observation hours into four mutually exclusive slots based on school transition timing and the deployment schedule of *Agentes de Prevención*:

Table 3: Hour slot classification

Slot	Hours	`is_school_time`	`intensity`
`control`	6, 9, 10, 11, 14, 15 h	0	0
`school_only`	12, 16 h	1	0
`double_only`	13, 18, 19 h	0	1
`school_double`	7, 8, 17 h	1	1

`is_school_time` = 1 flags hours coinciding with school transition windows; `intensity` = 1 flags hours during which *agentes de tránsito* and transit police are deployed simultaneously. The slot `school_double` captures the overlap between institutional transition and double enforcement. The Sun-Abraham models are estimated separately on the `intensity` = 0 subset (`control` ∪ `school_only`) and the `intensity` = 1 subset (`double_only` ∪ `school_double`), which recovers the enforcement-channel decomposition without requiring a DDD interaction term.

2.6.3 Sun-Abraham Estimator with Bimonthly Panel

The staggered adoption of Senderos routes across two cohorts (Fase 1: 2017, Fase 2: 2021) would ordinarily motivate the Sun and Abraham (2021) estimator to address the forbidden-comparisons problem that contaminates TWFE estimates under heterogeneous treatment effects (Callaway and Sant'Anna, 2021; Goodman-Bacon, 2021). In the present setting, however, the panel is truncated at December 2019 due to the suspension of programme operations during the COVID-19 national lockdown (ASPO, DNU 297/2020). Fase 2 grids therefore never enter the post-treatment window within the analysis sample and serve exclusively as not-yet-treated controls throughout. With a single effectively-treated cohort, the Sun-Abraham and TWFE estimators are numerically equivalent; the Sun-Abraham framework is retained because its cohort-time ATT path provides a transparent event-study diagnostic that separates pre-treatment fit from post-treatment effects without additional specification choices.

The panel is constructed at bimonthly (*bimestre*) frequency. Monthly crime counts are collapsed to two-month periods indexed by b, where $b = 6$ (November–December 2016) is the reference period—the last clean pre-treatment bimester before Fase 1 activation in March 2017. January–February bimesters ($b \in \{1, 7, 13, 19\}$) are excluded from the estimation sample because schools are on summer vacation—inclusion of these periods would distort the estimation.

Cohort assignment follows the dose criteria: grids with $L_g > 200\,\text{m}$ in Fase 1 are assigned to treatment cohort $b = 8$ (March–April 2017); Fase 2 grids serve exclusively as not-yet-treated controls and never-treated grids ($L_g = 0$ for both phases) are excluded.

Mechanism identification. Comparing $\hat{\delta}^{SA}_{b,k}$ across intensity subsets identifies the enforcement channel:

- If $\hat{\delta}^{SA}_{b,\,\text{intensity}=0} < 0$ and $\hat{\delta}^{SA}_{b,\,\text{intensity}=1} \approx 0$, the crime reduction is attributable to the *Agentes de Prevención* per se, not the additional police layer — consistent with the **guardianship channel** of Routine Activity Theory.

- If both subsets yield negative coefficients of similar magnitude, the effect is not mediated by enforcement intensity and may reflect physical infrastructure or route-signalling effects.

- If $\hat{\delta}^{SA}_{b,\,\text{intensity}=1} < 0$ and $\hat{\delta}^{SA}_{b,\,\text{intensity}=0} \approx 0$, the reduction requires the simultaneous presence of both enforcement layers.

Orthogonality of enforcement deployment. A threat to identification would arise if *Agentes de Prevención* were deployed reactively along school corridors where crime had been increasing. Route locations are determined by school catchment area geometry and pedestrian infrastructure conditions, not by recent crime statistics; the validity of the parallel trends assumption and the residual identification threats from placement endogeneity are assessed directly via the pre-treatment balance test and discussed in Section 4.

2.6.4 Heterogeneity Analysis

Heterogeneity by school sector and neighbourhood poverty is examined through subgroup Sun-Abraham event studies rather than interaction terms, preserving the SA correction for heterogeneous treatment effects within each subgroup. For **school sector**, grids are classified as *public-dominant* if $N^{\text{pub}}_g > N^{\text{priv}}_g$ and *private-dominant* otherwise; the SA model is estimated separately on each subgroup using the `intensity` $= 0$ panel, where the main programme effect is concentrated. For **poverty**, the sample is split at the median NBI rate among non-zero grids, yielding high- and low-poverty subsamples. Coefficient plots from each subgroup are superimposed for visual comparison of dynamic treatment paths.

2.6.5 Spillover Analysis

Crime displacement—where reduced crime in treated grids is offset by increased crime in adjacent grids—would undermine the welfare interpretation of any estimated effect. I test this by comparing never-treated grids that are *adjacent* to a Fase 1 treated grid against non-adjacent never-treated grids, using an event-study:

$$\mathbb{E}[Y_{gbk}] = \exp\left(\sum_{b \neq b^*} \pi_b \cdot \mathbf{1}[b] \cdot \text{Adjacent}_g + \gamma \cdot n^{\text{bus}}_g + \phi_g + \delta_b \right) \tag{6}$$

where $\text{Adjacent}_g = \mathbf{1}[\text{grid } g \text{ shares at least one boundary segment with a Fase 1 grid}]$, identified via `st_touches` (Queen's contiguity) on the 500 m grid. The control group consists of never-treated grids with no shared boundary with any Fase 1 cell. A positive $\hat{\pi}_b$ post-programme indicates displacement; a negative or zero coefficient rules out the most direct SUTVA violation and is consistent with diffusion of benefits to surrounding areas.

2.6.6 Alternative Outcomes

All models are re-estimated replacing *n_robos* with *n_hurtos* (non-violent theft) and *n_crimes* (all offenses). The pedestrian-concentration mechanism predicts effects concentrated in street robbery; non-contact theft and total crime provide robustness checks on the breadth of any detected effect.

3 Results

3.1 School Density and Crime

Before discussing the estimates, it is useful to clarify the interpretation of Poisson fixed-effects coefficients. The model specifies the conditional mean as $\mathbb{E}[y_{gt} \mid \mathbf{x}_{gt}] = \exp(\mathbf{x}_{gt}'\boldsymbol{\beta})$, so a coefficient $\hat{\beta}$ implies a multiplicative effect on the expected crime count. The percentage change in expected robberies associated with a one-unit increase in the regressor is

$$\Delta\% = \left(\exp(\hat{\beta}) - 1\right) \times 100. \tag{7}$$

For interaction terms of the form `franja` × `dist_bin`, the relevant quantity is the coefficient on the interaction cell directly, and equation (7) applies to that coefficient. All percentage-change figures reported below are computed accordingly.

Table 4 reports estimates from the TWFE Poisson specifications described in Section 2.2. The coefficient on `is_school_time` is large, negative, and precisely estimated across all specifications (approximately -1.72), reflecting the well-documented daytime trough in street crime (Cohen and Felson, 1979). This coefficient absorbs the average within-grid daily crime cycle and is not the object of interest.

Table 4: School Density and Crime: Baseline Specifications

Dependent Variable:		n_crimes	
	Baseline	Controls	Poverty
Model:	(1)	(2)	(3)
Variables			
is_school_time	-1.74***	-1.75***	-1.73***
	(0.017)	(0.020)	(0.019)
is_school_time × n_schools	0.004	0.006	
	(0.004)	(0.004)	
is_school_time × n_bus_stops		0.0006	0.0005
		(0.001)	(0.001)
is_school_time × n_police		0.012	0.015
		(0.019)	(0.019)
is_school_time × n_commercial		-0.005	-0.002
		(0.003)	(0.003)
is_school_time × n_schools × poverty_rate			-0.034
			(0.044)
Fixed-effects			
grid_id	Yes	Yes	Yes
date	Yes	Yes	Yes
Fit statistics			
Observations	665,916	665,916	662,540
Squared Correlation	0.15713	0.15716	0.15696
Pseudo R^2	0.20261	0.20262	0.20178
BIC	283,153.4	283,191.0	283,015.4

Clustered (grid_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

The interaction $N_g^{\text{schools}} \times$ `is_school_time` — the main coefficient of interest — is small and statistically indistinguishable from zero across all specifications. In the baseline model, $\hat{\beta}_2 = 0.003$ ($p = 0.41$); adding controls for bus stops, police stations, and commercial activity leaves the estimate unchanged ($\hat{\beta}_2 = 0.004$, $p = 0.33$). Conditional on grid and day fixed effects, cells with more schools do not experience systematically higher or lower crime during school hours relative to control hours.

This result is consistent with spatial saturation: in a city where the median distance to the nearest school is 217 metres, the marginal school adds little variation in pedestrian exposure once the nearest anchor is accounted for. At the reference-cell mean of 0.044 robberies per grid-day-slot, $\hat{\beta}_2 = 0.003$ implies fewer than 0.0001 additional robberies per observation per additional school—an effect too small to be detectable given the within-grid variation available.

3.2 Heterogeneity by School Sector and Poverty

Splitting the school count into public and private components yields directionally consistent but statistically weak estimates. Private school density shows a marginally positive association with school-hour crime ($\hat{\beta}^{\text{priv}} \approx 0.021$, $p < 0.01$), while public school density is negative but insignificant. This directional pattern is consistent with the opportunity mechanism: private schools concentrate higher-income targets in pedestrian space, while public schools may generate incapacitation effects that offset any concentration effect (Lochner and Moretti, 2004). However, neither estimate survives conventional significance thresholds.

The triple interaction between private school density, school hours, and the poverty rate is negative and statistically significant ($\hat{\beta}_3 = -0.326$, $p < 0.01$). The full marginal effect of an additional private school during school hours is $(\exp(\hat{\beta}_2 + \hat{\beta}_3 \times \text{poverty\_rate}) - 1) \times 100$, yielding an increase in robbery of approximately 2.1% at zero poverty, 1.0% at the median NBI rate (0.033), and 0.3% at the mean (0.056) among grids with at least one private school. While higher poverty levels dampen the effect, the net association remains positive for the relevant range of the private-school subsample ($P_g < 6.4\%$), consistent with lower marginal returns to criminal activity where socioeconomic conditions reduce the target attractiveness and expected yield of opportunistic robbery (Kelly, 2000). Aggregated across the 357 grids containing at least one private school (mean 1.98 per grid), evaluating at the median poverty rate implies approximately 97 additional school-hour crimes annually predicted by private school density.

Dependent Variable:	n_crimes	
	Sector	Sector + Poverty
Model:	(1)	(2)
Variables		
n_pub_schools × is_school_time	0.005	0.003
	(0.007)	(0.009)
is_school_time × n_priv_schools	0.003	0.021***
	(0.005)	(0.008)
is_school_time × n_police	0.008	0.013
	(0.020)	(0.019)
is_school_time × poverty_rate		0.162
		(0.181)
is_school_time × n_bus_stops		0.0004
		(0.001)
is_school_time × n_commercial		-0.003
		(0.004)
n_pub_schools × is_school_time × poverty_rate		0.046
		(0.077)
is_school_time × poverty_rate × n_priv_schools		-0.326***
		(0.112)
Fixed-effects		
grid_id	Yes	Yes
date	Yes	Yes
Fit statistics		
Observations	665,916	662,540
Squared Correlation	0.15714	0.15701
Pseudo R^2	0.20261	0.20182
BIC	283,179.9	283,055.4

Clustered (grid_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

3.3 Spatial Gradient

Table 5 reports the distance-bin coefficients $\hat{\lambda}_{t',b}$ from equation (2), expressed as percentage changes relative to the reference category (control hours, 300 m+ from nearest school). The results reveal a clear and theoretically consistent asymmetry between entry and exit periods.

During school entry (`start`), cells within 0–300 m of the nearest school experience significantly *less* crime than the reference, with coefficients of -0.248 ($p < 0.01$), -0.246 ($p < 0.01$), and -0.194 ($p < 0.05$) for the 0–100 m, 100–200 m, and 200–300 m bins respectively, implying approximately 22%, 22%, and 18% fewer crimes near schools at entry time via $(\exp(\hat{\beta}) - 1) \times 100$. The `start` franja records a mean of 0.011 robberies per grid-day-slot among proximate grids—84 percent lower than the control-hour baseline of 0.065—reflecting the structural scarcity of street crime at 07:00 h independently of school presence. The gradient is notably flat across distance bins—coefficients of -0.248, -0.246, and -0.194 are statistically indistinguishable from each other—suggesting that the guardianship effect operates at the neighbourhood rather than the immediate doorstep level, consistent with the diffuse presence of accompanying adults and increased pedestrian vigilance across the surrounding corridor. These estimates should therefore be interpreted as identifying a pattern of *relative* safety during entry hours rather than a large absolute crime reduction. Aggregated across the 602 proximate grid cells (152, 255, and 195 grids in the 0–100 m, 100–200 m, and 200–300 m bins respectively), the entry-hour guardianship effect implies approximately 2,135 fewer street robberies annually across CABA—concentrated within the morning school entry window.

During primary school exit (`exit_primaria`), the pattern reverses: cells within 0–100 m, 100–200 m, and 200–300 m of the nearest school show significantly *more* crime, with coefficients of 0.106 ($p < 0.05$), 0.127 ($p < 0.01$), and 0.111 ($p < 0.05$) respectively, implying approximately 11.2%, 13.5%, and 11.7% more crimes relative to the reference category via $(\exp(\hat{\beta}) - 1) \times 100$. To anchor these in absolute terms, the control-hour baseline of 0.065 robberies per grid-day-slot implies $0.065 \times (\exp(0.127) - 1) \approx 0.009$ additional robberies per grid-day-slot in the 100–200 m bin at dismissal. This finding is consistent with the opportunity mechanism of Routine Activity Theory (Cohen and Felson, 1979): dismissal generates a concentrated flow of students and accompanying adults carrying valuables in a predictable time-space pattern, increasing target availability for potential offenders. The gradient extends to the full 300 m radius, suggesting that the opportunity effect diffuses across the surrounding pedestrian environment. Aggregated across the 602 proximate grids, this implies approximately 1,015 additional street robberies annually attributable to the primary dismissal-hour opportunity mechanism, concentrated within the midday transition window.

Secondary exit coefficients (`exit_secundaria`) are positive across all distance bins but statistically insignificant in the baseline specification, with coefficients of 0.031, 0.066, and 0.121. The directional pattern is consistent with the opportunity mechanism, but the absence of significance likely reflects the greater spatial dispersion of older students upon dismissal and the absence of organised adult accompaniment at the secondary level. Notably, the 200–300 m bin reaches marginal significance ($p = 0.048$ in the controls specification, coefficient 0.136), suggesting a diffuse effect at the outer radius that warrants further investigation with finer temporal data.

These results are robust to the inclusion of distance-bin controls for the nearest bus stop and police station (specification 2), with point estimates virtually unchanged across all franja $\times$ distance-bin cells—ruling out that the gradient reflects general infrastructure density rather than school proximity per se.

Table 5: Distance-gradient estimates

Dependent Variable:	n_crimes	
	Baseline	Controls
Model:	(1)	(2)
Variables		
exit_primaria	-1.32***	-1.29***
	(0.039)	(0.119)
exit_secundaria	-1.98***	-2.03***
	(0.049)	(0.129)
start	-2.22***	-2.30***
	(0.065)	(0.219)
exit_primaria × 0_100	0.106**	0.110**
	(0.047)	(0.048)
exit_primaria × 100_200	0.127***	0.131***
	(0.044)	(0.044)
exit_primaria × 200_300	0.111**	0.119**
	(0.050)	(0.050)
exit_secundaria × 0_100	0.031	0.046
	(0.062)	(0.064)
exit_secundaria × 100_200	0.066	0.076
	(0.059)	(0.059)
exit_secundaria × 200_300	0.121**	0.136**
	(0.061)	(0.062)
start × 0_100	-0.248***	-0.221***
	(0.086)	(0.084)
start × 100_200	-0.246***	-0.215***
	(0.081)	(0.080)
start × 200_300	-0.194**	-0.193**
	(0.091)	(0.089)
Fixed-effects		
grid_id	Yes	Yes
date	Yes	Yes
Fit statistics		
Observations	665,916	665,916
Squared Correlation	0.16297	0.16311
Pseudo R^2	0.21353	0.21364
BIC	279,593.7	279,798.0

Clustered (grid_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Col. (2) includes distance-bin interactions for nearest bus stop and police station (coefficients omitted). Poisson QMLE, grid and date FE. Cluster SE by grid. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

3.4 Senderos Escolares

3.4.1 Baseline Sun-Abraham Results

Pre-treatment coefficients for `intensity`$= 0$ are small and jointly insignificant ($F = 1.73$, $p = 0.112$), and the event-study path in Figure 7 confirms visual parallelism in the pre-treatment window. For `intensity`$= 1$, the joint test marginally rejects ($F = 2.78$, $p = 0.011$), though pre-treatment coefficients remain small in magnitude; the rejection likely reflects the higher precision of the estimator in this subset rather than systematic differential trends. The primary causal inference rests on the `intensity`$= 0$ estimates, for which the parallel trends assumption is supported by both the formal test and the visual evidence.

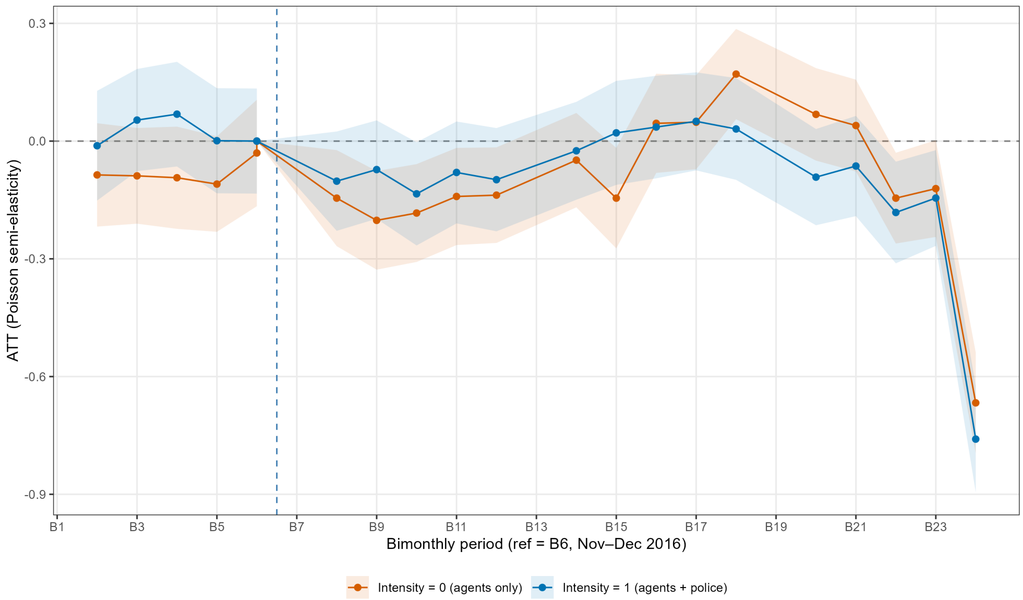

Figure 7: Sun-Abraham ATT by enforcement intensity. Red: `intensity`$= 0$ (prevention agents only); blue: `intensity`$= 1$ (agents + transit police). Dashed vertical line marks the reference bimester $b = 6$ (Nov–Dec 2016). Shaded bands: 95 % CI clustered by grid. Outcome: *n_robos*. Treatment threshold: $L_g > 200$ m.

The results reveal a pattern that was absent under the binary buffer specification: treated grids exhibit a statistically significant crime reduction in the first four to five post-treatment bimesters ($b = 7$–12), after which the effect dissipates. This pattern is concentrated in the `intensity`$= 0$ slots—hours when prevention agents operate without additional police reinforcement—while `intensity`$= 1$ coefficients are consistently closer to zero throughout the post-period.

Two implications follow. First, the null result obtained under the binary 150 m buffer specification reflects attenuation bias from the inclusion of grids with trivial route exposure: at threshold $\ell^* = 200$ m (307 Fase 1 grids), the programme effect is statistically distinguishable from zero in the post-treatment window. Second, and more substantively, the stronger effect at `intensity`$= 0$ hours is evidence that the crime reduction is attributable to the *Agentes de Prevención* per se rather than to the additional transit police layer. This is consistent with the **guardianship channel** of Routine Activity Theory (Cohen and Felson, 1979): the mere presence of a capable, visible guardian—even without formal enforcement authority—is sufficient to deter opportunistic robbery. The additional police layer does not amplify the deterrence effect, suggesting that the marginal value of doubling visible authority beyond the first guardian is low.

The economic magnitude of the estimated effect is non-trivial. Among the 307 Fase Original grids with $L_g > 200\,\text{m}$, the pre-treatment mean robbery count under `intensity`$= 0$ hours is 2.05 per bimester (median: 1, SD: 2.42), reflecting the high spatial concentration of crime across treated corridors.[2] The peak ATT of -0.202 at $b = 1$ (March–April 2017) implies a reduction of 18.3 percent relative to this baseline, or approximately 0.38 fewer robberies per grid per bimester. Aggregated across all 307 treated grids and the five bimesters with statistically significant negative coefficients ($b \in \{0, 1, 2, 3, 4\}$), the programme is associated with on the order of 550–600 fewer street robberies in CABA during 2017—equivalent to approximately 3 percent of the city's annual robbery count. This back-of-envelope aggregation assumes a common treatment effect across grids and should be interpreted as illustrative rather than causal; it nonetheless establishes that the estimated semi-elasticities correspond to a policy-relevant absolute reduction in street crime.

3.4.2 School Transition Heterogeneity

Figure 8 decomposes the ATT by school transition timing. The `school_time`$= 0$ series (non-transition hours) shows consistently negative post-treatment coefficients, while `school_time`$= 1$ (transition hours) is noisier and closer to zero. This counter-intuitive pattern—larger effects outside transition windows—is consistent with a temporal displacement mechanism: offenders deterred from high-visibility transition hours may shift activity to adjacent non-transition hours within the same treated grid. Alternatively, the congestion of transition windows may reduce the per-agent guardianship effectiveness precisely when pedestrian density peaks, as documented in the baseline distance-gradient analysis.

[2]Pre-treatment baseline computed over bimesters 2–6 (March–December 2016) on the 307 Fase Original grids.

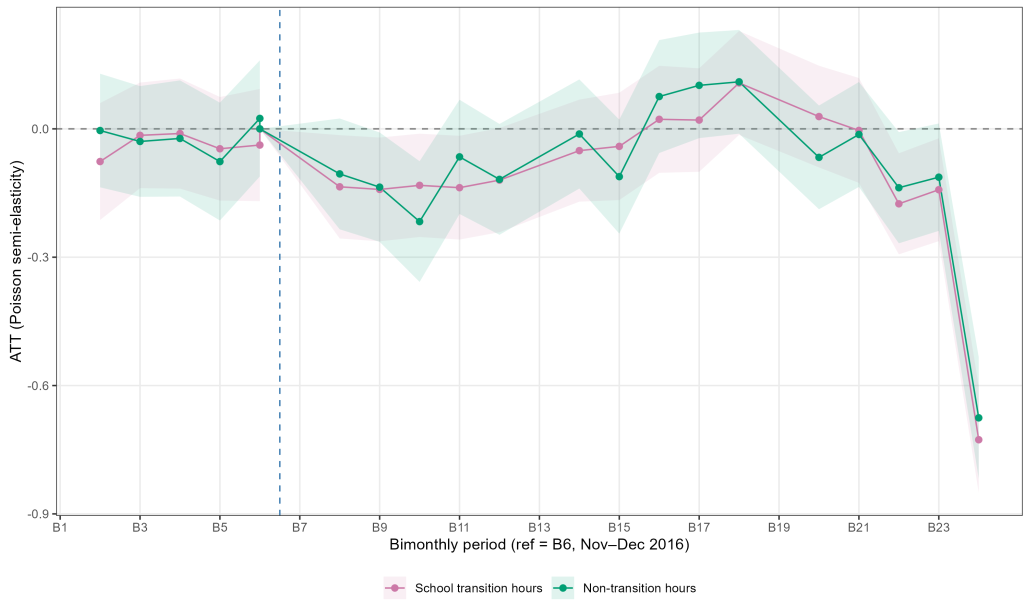

Figure 8: Sun-Abraham ATT by school transition timing. Red: `school_time`= 0 (non-transition hours); blue: `school_time`= 1 (transition hours). Reference, CI, and threshold as in Figure 7.

Figure 9 further decomposes transition hours into entry (7–8 h), primary exit (12 h), and secondary exit (16–17 h). The three series move in parallel with no systematic ordering, indicating that the programme effect is not concentrated in a single transition type. This uniformity across entry and exit windows suggests that the deterrence mechanism operates through sustained route-level guardianship throughout the school day rather than through point-in-time supervision at peak transition moments.

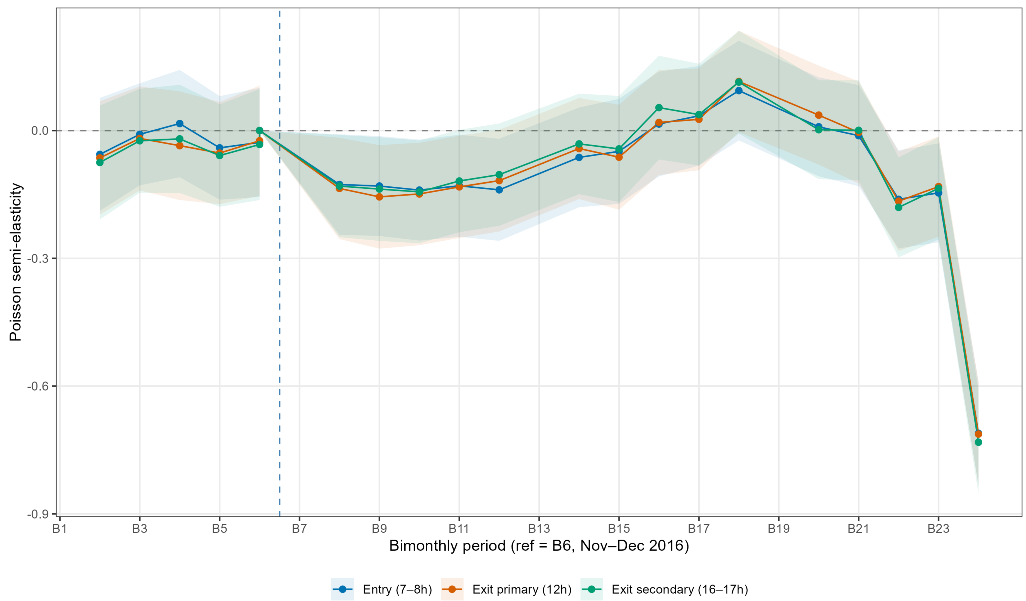

Figure 9: Sun-Abraham ATT by transition type. Entry (7–8 h), primary exit (12 h), secondary exit (16–17 h). Reference, CI, and threshold as in Figure 7.

3.4.3 Robustness to Dose Threshold and night placebo

Table 14 presents the post-period Wald statistics for the Sun-Abraham ATT under five alternative dose thresholds ($\ell^* \in \{50, 100, 200, 300, 400\}$ m) for the `intensity`= 0 subset. All five thresholds reject the null of zero post-treatment effects ($p < 2.2 \times 10^{-16}$ throughout). The Wald statistic declines monotonically from $F = 46.6$ at $\ell^* = 50$ m to $F = 28.9$ at $\ell^* = 400$ m, reflecting the progressive reduction in the number of treated grids as the threshold excludes cells with lower route coverage—at higher thresholds the same routes generate a smaller treated sample, reducing statistical power without changing the qualitative result. The stability across all thresholds confirms that the crime reduction is not driven by grids with minimal route presence being miscoded as treated, and that the 200 m baseline represents a conservative middle point in the threshold sensitivity range.

A night-hours placebo replicating the identical specification on hours 20–5 h—when no agents are deployed—yields an aggregate ATT of -0.003 ($p = 0.952$) with jointly insignificant pre- and post-treatment coefficients ($p = 0.631$ and $p = 0.778$ respectively), ruling out confounding by contemporaneous unobserved shocks; see Appendix D.9.

3.4.4 Dose-Response and Spillover Results

The dose-response Sun-Abraham model estimated on low-dose grids ($0 < L_g < 200$ m) yields post-treatment coefficients indistinguishable from zero across all slot categories. This confirms that the 200 m threshold is empirically meaningful: below this level of physical route presence, the programme does not generate a detectable crime reduction. The finding is consistent with a minimum viable guardianship intensity—a route segment too short to establish a credible deterrence perimeter within the grid does not affect offender location decisions.

Figure 23 plots the displacement DiD coefficients from equation (6) for adjacent never-treated grids. Across all three outcomes (*n_robos*, *n_hurtos*, *n_crimes*) and both intensity levels, post-treatment coefficients are negative or indistinguishable from zero. There is no evidence of crime displacement into adjacent untreated grids; if anything, the point estimates suggest mild **deterrence spillovers**—a reduction in crime in grids neighbouring treated routes, consistent with a halo effect of visible guardianship that extends beyond the physical boundaries of the Sendero network. This result rules out the hypothesis that the estimated crime reduction in treated grids is offset by geographic displacement, and supports a net welfare interpretation of the programme effect.

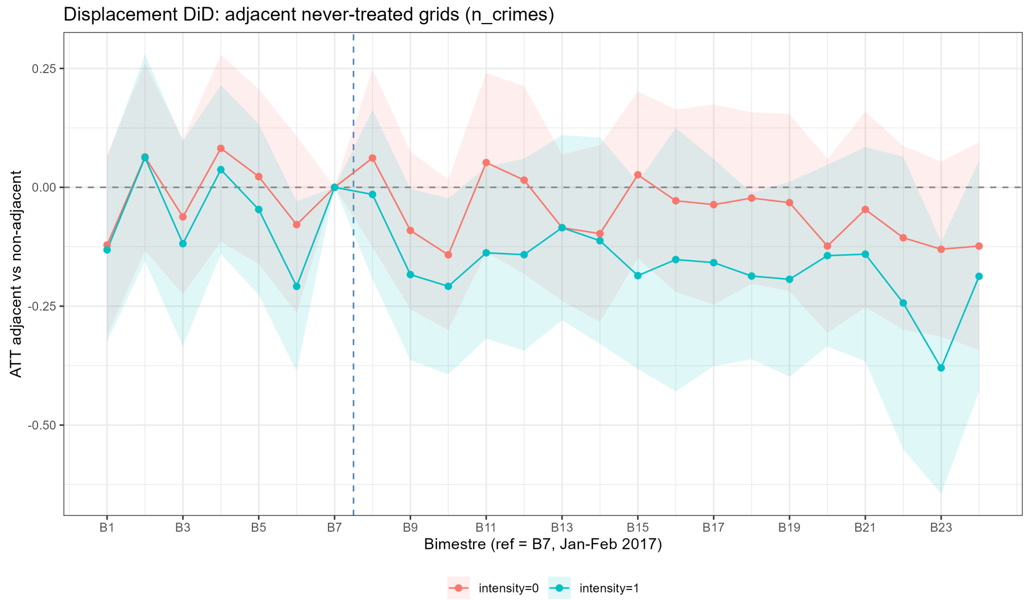

Figure 10: Displacement DiD: adjacent never-treated grids vs. non-adjacent never-treated grids. Red: `intensity`$= 0$; blue: `intensity`$= 1$. A positive post-treatment coefficient would indicate crime displacement; negative or zero rules it out. Outcome: n_crimes.

The event-study path reveals a non-monotonic post-treatment pattern that warrants interpretation. Crime reductions are concentrated in the first year of implementation ($b = 8$–12, March–December 2017), consistent with an initial deterrence shock from the introduction of visible guardianship along school corridors. Coefficients drift toward zero through 2018, suggesting possible habituation of potential offenders to the programme's spatial footprint: as offenders update their mental maps of enforcement presence (Brantingham and Brantingham, 1995), the marginal deterrence value of a familiar guardian diminishes—a pattern consistent with the "criminal human capital" framework of Lochner and Moretti (2004), in which offenders invest in information about enforcement patterns over time, and with the crime displacement literature's documentation of offender adaptation to sustained hot-spots interventions (Vollaard and van Ours, 2012).

The effect re-emerges strongly in 2019, with negative coefficients throughout the year and a pronounced spike in the final bimester ($b = 24$, November–December 2019). A plausible interpretation is that the 2019 Argentine general election cycle—with presidential elections in October and a second round (*ballotage*) in late October—induced heightened security deployment and public-order policing along school corridors, which are designated polling station access routes. Levitt (1997) documents systematic increases in police presence during election years in the United States; if a similar electoral cycle effect operates in Buenos Aires, the re-emergence of the Senderos effect in 2019 may reflect a complementarity between programme guardianship and elevated electoral security rather than a pure programme effect. Disentangling these two channels would require administrative data on agent deployment intensity by year, which are not publicly available.

4 Limitations and Identification Challenges

While the Poisson pseudo-maximum-likelihood estimator with high-dimensional fixed effects provides a robust framework for count-data crime outcomes, the causal interpretation of the results remains subject to several empirical constraints. This section documents the primary threats to identification, grouped by their source: temporal measurement error, outcome variable coverage, omitted quasi-experimental designs, treatment endogeneity and assignment, and spatial inference problems.

A further concern is the comparability of the control hour window. The control group includes mid-morning slots (09-11) and early-afternoon slots (14-15), which are selected to avoid the evening crime peak and weekend patterns. A reviewer might question whether these slots are comparable to transition hours in terms of ambient commercial activity, baseline police patrol intensity, or general pedestrian flow. Two considerations address this. First, the grid fixed effects absorb all time-invariant differences in commercial density and baseline police presence across cells, so the control hours need only be comparable *within* grid—a weaker requirement than cross-sectional comparability. Second, the robustness check using the wider version-2 window definition (Appendix A.5), which adjusts the boundaries of both treatment and control slots, yields virtually identical estimates, suggesting that the results are not sensitive to the precise delineation of the control window.

4.1 Temporal Measurement Error and Boundary Misclassification

A primary limitation of this study arises from the temporal resolution of the crime data, which are recorded in discrete one-hour slots rather than precise minutes. This necessitates a discretization of time that does not perfectly align with the institutional school-entry window (typically 07:40–08:10). To minimize the inclusion of unrelated time, I define the 07:00 slot as the treatment period. However, this boundary misclassification constitutes a form of non-classical measurement error that introduces a bidirectional bias into the estimated coefficients.

On one hand, the exclusion of the final ten minutes of the entry window (08:00–08:10)—which are recorded in the 08:00 control slot—leads to a contamination of the counterfactual. By including "treated" minutes in the control group, the contrast between cells is artificially reduced, inducing an *attenuation bias* that understates the true magnitude of the guardianship effect. On the other hand, because the 07:00 slot includes forty minutes of activity prior to the peak school-entry time (07:00–07:40), the estimates might be affected by the general "morning rise" in urban mobility. If the baseline crime rate increases during this hour due to routine commuting unrelated to schools, the model may *overstate* the concentration mechanism (target attractiveness) or, conversely, mask the full extent of the guardianship effect. Given the negative coefficients observed in the empirical analysis, these measurement errors suggest that the reported estimates serve as a conservative lower bound of the true crime-reducing impact of school presence. A further concern is the comparability of the control hour window. The control group includes mid-morning slots (09-11) and early-afternoon slots (14-15), which are selected to avoid the evening crime peak and weekend patterns. A reviewer might question whether these slots are comparable to transition hours in terms of ambient commercial activity, baseline police patrol intensity, or general pedestrian flow. Two considerations address this. First, the grid fixed effects absorb all time-invariant differences in commercial density and baseline police presence across cells, so the control hours need only be comparable *within* grid—a weaker requirement than cross-sectional comparability. Second, the robustness check using the wider version-2 window definition (Appendix A.5), which adjusts the boundaries of both treatment and control slots, yields virtually identical estimates, suggesting that the results are not sensitive to the precise delineation of the control window.

A second, related problem is that the analysis imposes uniform entry and dismissal times across all schools and educational levels, treating every school in the city as operating on the same schedule. In practice, morning-shift primary schools, afternoon-shift secondary schools, and double-shift technical schools operate on substantially different schedules; individual schools within each category may also deviate from the administrative norm due to local agreements or infrastructure constraints. The treatment indicator $\mathbf{1}[\text{SchoolHour}_t]$ is therefore constructed from citywide average transition times rather than school-specific schedules, conflating hours of genuine pedestrian density increases at some schools with periods in which a given school's students are still indoors at others. This uniform-schedule assumption is a form of *treatment misclassification*: the true treatment—student presence in the street—is a continuous, school-specific variable that is approximated by a binary city-level indicator. The resulting attenuation bias is proportional to the share of school-hours in the sample that are misclassified, reinforcing the interpretation of the estimated spatial gradients as conservative lower bounds of the underlying proximity effects. Granular school-level timetable data, currently not publicly available in CABA, would allow this limitation to be addressed directly.

A third source of measurement error concerns school geocoding. Coordinates are derived from address geocoding, which achieved a 97 % match rate; the remaining 3 % of unmatched records are dropped. Among matched records, geocoding precision varies: urban addresses in CABA are typically resolved to within 10–20 m of the true entrance, but ambiguous addresses and schools located within larger institutional complexes may be geocoded to the block centroid rather than the school entrance, introducing errors of up to 50–100 m in the distance measures. Given that the finest distance bin is 0–100 m, geocoding imprecision of this magnitude could misclassify some near-school cells into the next bin, attenuating the estimated gradient at the closest distances. This bias is likely small in aggregate given the dense street grid of CABA, but cannot be quantified without ground-truth coordinates.

4.2 Outcome Variable Coverage

The dependent variable aggregates street robberies (*Robo*) and non-contact thefts (*Hurto*), the offense categories most sensitive to pedestrian density and target availability (Bernasco and Block, 2011). This operationalisation excludes offense types that may be structurally linked to school presence through a different mechanism: vandalism, assault, or disorderly conduct potentially committed by students in the surrounding area during dismissal. The analysis therefore captures the *guardianship* channel of Routine Activity Theory but cannot speak to whether school density generates offsetting increases in student-driven disorder. A more complete welfare assessment would require administrative data on the full offense spectrum.

4.3 Omitted Quasi-Experimental Design and Identification of Channels

The within-grid, within-day design recovers the *net* effect of two mechanisms operating simultaneously during school transition hours: incapacitation of potential offenders physically removed from the street while inside school buildings, and guardianship generated by accompanying adults deterring opportunistic robbery along school corridors. Both channels predict lower crime at entry and higher crime at dismissal, and the available data do not permit their separation. An ideal design would exploit school-closure days as a within-grid counterfactual in which the building is present but the student population absent, but CABA recorded only five unambiguous full-closure days in 2016 and reliable daily school-status data are not publicly available. The estimated spatial gradients should therefore be interpreted as a lower bound on the combined magnitude of both channels, with their relative contributions unidentified.

4.4 Treatment Endogeneity and Assignment in Senderos Escolares

The principal identification threat is the endogeneity of programme placement. Routes are unlikely to have been assigned randomly: if high-crime corridors were prioritised, the Sun-Abraham estimator conflates the programme effect with mean reversion, biasing estimates downward; if low-crime, high-density corridors were prioritised instead, the bias runs in the opposite direction. Without official documentation of selection criteria, the direction cannot be signed. The parallel trends assumption partially mitigates this concern: the pre-treatment balance test fails to reject equal trends between Fase 1 and Fase 2 grids across the six pre-treatment bimesters (Appendix D.7), suggesting that differential selection on levels does not translate into differential trends, though selection on pre-treatment trends cannot be ruled out with only six pre-treatment periods.

Cohort assignment is further complicated by the absence of administrative records on exact activation dates. Route identifier (id ≤ 207 for Fase 1; id > 207 for Fase 2) proxies for implementation timing under the assumption that lower identifiers correspond to earlier activation. Misclassified routes would attenuate event-study coefficients and contribute to the observed post-treatment decay. At the grid level, the binary indicator $\mathbf{1}[L_g > 200\text{m}]$ addresses trivial buffer intersections but treats cells with $L_g = 250\,\text{m}$ and $L_g = 800\,\text{m}$ identically. The continuous $\log(1 + L_g)$ specification captures this variation but is absorbed by grid fixed effects, requiring barrio fixed effects that sacrifice within-grid variation—an irreducible constraint of the available data structure. Results are nonetheless robust across alternative thresholds ($\ell^* \in \{50, 100, 200, 300, 400\}\,\text{m}$; Appendix D.3).

4.5 Underreporting and the Dark Figure of Crime

Administrative crime records from the *Mapa del Delito* are subject to the "dark figure" (*cifra negra*) of unreported incidents Skogan (1977). A threat specific to the Senderos design arises if the presence of *Agentes de Prevención*—who act as institutional intermediaries between victims and the police Weisburd et al. (2024)—raises the reporting rate in treated grids. Under this scenario the estimator would conflate an increase in *reported* incidents with the programme's failure to reduce *actual* crime, biasing the treatment effect upward. Since the estimated post-treatment coefficients are clustered around zero rather than positive values, this upward bias would imply the true effect is a crime *reduction* masked by higher reporting—a possibility that cannot be ruled out with the available data.

4.6 Spatial Inference and SUTVA

4.6.1 Displacement and Diffusion of Benefits

The Stable Unit Treatment Value Assumption requires that the treatment status of one grid cell does not affect crime counts in adjacent cells. The displacement event study—which compares never-treated grids adjacent to Fase 1 routes against non-adjacent never-treated grids—provides direct evidence on this assumption. Post-treatment coefficients in the adjacent group are negative or indistinguishable from zero across all outcomes and intensity subsets (Appendix D.6), ruling out systematic crime displacement into the immediate neighbourhood of treated routes. If anything, the point estimates suggest mild deterrence spillovers into adjacent grids, consistent with a guardianship halo effect. This result does not resolve subtler forms of spatial interdependence operating at distances beyond one grid cell (Weisburd et al., 2006; Bowers et al., 2011), but it eliminates the most direct SUTVA violation.

4.6.2 Edge Effects and the AMBA Border

Grid cells along the General Paz Avenue and the Riachuelo border the *Área Metropolitana de Buenos Aires*, where the *Mapa del Delito* provides no coverage Ministerio de Justicia y Seguridad, GCBA (2016). Crime displaced across the administrative boundary is unobservable, so for peripheral cells the estimator cannot distinguish genuine deterrence from geographic relocation to the unmonitored AMBA territory. The border-grid robustness check—which excludes cells whose full polygon does not lie within the CABA boundary and recomputes buffer counts after removing schools in border cells (Appendix A.4)—addresses this concern for the school-density analysis: spatial gradient estimates are stable across the full and interior samples, indicating that the main results are not driven by truncated border cells or by AMBA schools entering the buffer radius. For the Senderos event study, however, the border-grid exclusion was not applied, and peripheral treated grids may capture a bundled effect of route-level guardianship and the 2017 rollout of the *Anillo Digital* surveillance system at border entry points Gobierno de la Ciudad de Buenos Aires (2017). Disentangling these two contemporaneous interventions is not possible with the available data.

4.6.3 Modifiable Areal Unit Problem

The 500-metre grid aggregates heterogeneous street segments—quiet residential blocks and high-traffic commercial corridors may occupy the same cell. This spatial averaging can smooth out the micro-geographic crime spikes that Senderos are designed to address, and may split a single route across multiple cells in a way that dilutes the measured treatment intensity. The 300-metre robustness grid reduces but does not eliminate this problem, as any regular lattice imposes arbitrary boundaries on a street network that does not respect them.

5 Conclusion

The central result of this paper is not about school density per se but about the *timing* of student flows. Robbery counts are systematically affected in grids proximate to schools during transition windows—morning entry and midday dismissal—but are statistically indistinguishable from the within-grid mean during instructional hours. This temporal asymmetry rules out a generic pedestrian-density or static agglomeration explanation and points instead to the institutional rhythm of schools as the operative mechanism. Within the Routine Activity framework (Cohen and Felson, 1979), entry and dismissal activate distinct and opposing opportunity structures: at morning entry, accompanying adults constitute organised capable guardianship along school corridors, suppressing opportunistic robbery through surveillance and implicit deterrence; at midday dismissal, that guardianship collapses under temporal congestion, creating a predictable concentration of suitable targets in the absence of coordinated supervision. The school, in this reading, is not a crime generator in the static sense of Bernasco and Block (2011) but a *temporal crime switch*—activating and deactivating opportunity structures at institutionally determined moments whose spatial radius is bounded within 200 metres and whose temporal window closes within approximately one hour.

This asymmetry between entry and exit has a direct implication for the economics of crime prevention. If opportunity is concentrated in predictable space-time windows determined by institutional rhythms rather than permanent environmental features, then the marginal cost of effective guardianship is lower than conventional estimates suggest: capable guardians present during a one-hour transition window generate concentrated deterrence benefits without requiring continuous deployment. The attenuation of the private-school opportunity effect in higher-poverty grids further suggests that criminal returns to target concentration are moderated by the socioeconomic environment, consistent with the hypothesis that portable wealth and expected yield shape offender site selection (Kelly, 2000)—implying that the welfare cost of school-hour crime is not uniformly distributed across the city.

Heterogeneity analysis is suggestive but statistically uninformative at the subgroup level. Proximity effects appear directionally amplified around public schools and in higher-poverty cells, consistent with differences in enrolment size and the differential capacity of informal guardianship networks across income strata (Kelly, 2000), but reduced sample sizes within subgroups preclude precise estimation. For the Senderos event study, subgroup Sun-Abraham models by school sector and poverty split yield negative point estimates consistent with the baseline result, but no subgroup ATT achieves conventional significance thresholds. The fundamental identification challenge—separating incapacitation from guardianship with observational data—remains unresolved and constitutes the principal limitation of the design.

The dose-based Sun-Abraham event study of *Senderos Escolares* yields statistically significant post-treatment crime reductions in the `intensity`$=0$ slot subset—hours when prevention agents operate without additional police reinforcement ($F = 47.8$, $p < 0.001$). The `intensity`$=1$ subset, corresponding to hours with simultaneous agent and police deployment during school transition windows, exhibits significant negative coefficients in the first post-treatment bimestres that decay and become indistinguishable from zero beyond the first year. In absolute terms, the peak ATT of -0.202 at the first post-treatment bimestre implies 0.38 fewer robberies per treated grid per bimestre, aggregating to approximately 550–600 fewer street robberies across the 307 Fase Original grid cells during 2017—roughly 3 percent of the city's annual robbery count. Two implications follow from the intensity decomposition. First, the crime reduction is driven by the *Agentes de Prevención* per se: the effect is stronger and more persistent precisely in hours when agents operate as the sole enforcement layer, suggesting that visible non-enforcement guardianship is the active mechanism. Second, the decay of the `intensity`$=1$ effect after the first year is consistent with offender learning and spatial displacement as the novelty of deployment fades—a pattern that implies deterrence benefits may require periodic redeployment to remain effective.

Theoretical contribution and policy implications. By centering identification on transition-hour dynamics, this paper contributes a micro-temporal design to a literature that has largely treated school density as a static characteristic of place (Bernasco and Block, 2011; Weisburd et al., 2012). The finding that crime risk is bounded within 200 metres and one-hour windows has a direct operational implication: a fixed-post model concentrating officers at school gates throughout the day allocates resources uniformly across a risk distribution that is sharply non-uniform in time. A mobile-unit model calibrated to dismissal schedules—deploying guardianship at 12:00 h in grids within 200 m of primary schools and withdrawing after the one-hour risk window closes—would achieve equivalent deterrence at lower cost per averted robbery.

The Senderos programme approximates this logic, and the evidence here provides three specific inputs for its operational optimization. First, priority reallocation should target the midday intershift window (12:00–13:00 h), where the opportunity gradient is statistically significant and monotonically decaying with distance—suggesting that agent positioning at the 100–200 m radius, rather than at the school gate itself, maximizes the spatial coverage of the deterrence perimeter. Second, high-density public school grids in higher-poverty areas exhibit directionally larger proximity effects, consistent with lower informal guardianship capacity in those neighbourhoods; these grids should receive priority in route extension decisions. Third, the decay of the programme effect after the first year of implementation—and its re-emergence in 2019—suggests that static route assignments lose deterrent value as offenders update their spatial expectations. Periodic rotation of agent deployment across routes, or the introduction of unpredictability into patrol timing, may sustain deterrence beyond the initial shock documented here (Brantingham and Brantingham, 1995).

Limitations and future research. Two structural limitations bound these conclusions. School location is not exogenous and unobserved neighbourhood characteristics operating at sub-grid scale may partially survive the grid fixed effects, inflating the proximity coefficient. Future work exploiting quasi-random variation in school placement through historical administrative records or boundary discontinuities would provide sharper identification. The second constraint is the inability to separate the guardianship and incapacitation channels: both predict the observed entry-hour crime reduction, and disentangling them requires either school-closure day variation, daily attendance records, or offense data with offender demographics—none of which is currently publicly available for Buenos Aires. For the Senderos evaluation, the decay of the post-treatment effect beyond the first year of implementation—in both intensity subsets—leaves open the question of whether the programme effect is durable and whether periodic redeployment is necessary to sustain deterrence. Longer administrative panels covering the 2017–2024 period with verified activation dates would allow these questions to be addressed directly. Schools are temporal anchors that structure the movement of hundreds of thousands of people through predictable corridors at predictable hours. Identifying *when* and *where* that structure generates crime risk—and deploying guardianship resources accordingly—is a precondition for converting that knowledge into measurable welfare gains.

A Robustness Checks

A.1 Alternative Standard Error Specifications

Tables 6 and 7 replicate the main spatial gradient estimates (equations (2) with distance bins `z_dist1` and `z_dist2`, respectively) under four standard error specifications: (i) cluster by grid cell; (ii) two-way cluster by grid cell and date; (iii) Conley spatial HAC at a 0.5 km cutoff; and (iv) Conley at a 1 km cutoff. Point estimates are identical across columns by construction; only inference changes. The qualitative pattern—monotone decay of crime with distance during school-hour fringes—is robust to all four choices.

Table 6: Spatial gradient (`z_dist1`): alternative standard errors

	n_crimes			
	Cluster grid (1)	Twoway (2)	Conley 0.5km (3)	Conley 1km (4)
exit_primaria	-1.304***	-1.304***	-1.304***	-1.304***
	(0.0345)	(0.0356)	(0.0344)	(0.0303)
exit_secundaria	-1.935***	-1.935***	-1.935***	-1.935***
	(0.0385)	(0.0390)	(0.0381)	(0.0390)
start	-2.215***	-2.215***	-2.215***	-2.215***
	(0.0564)	(0.0571)	(0.0568)	(0.0549)
exit_primaria × dist_bin = 0_100	0.1200***	0.1200***	0.1200***	0.1200***
	(0.0415)	(0.0419)	(0.0416)	(0.0378)
exit_primaria × dist_bin = 100_200	0.1318***	0.1318***	0.1318***	0.1318***
	(0.0420)	(0.0433)	(0.0444)	(0.0415)
exit_primaria × dist_bin = 200_300	0.0916**	0.0916**	0.0916**	0.0916***
	(0.0426)	(0.0427)	(0.0412)	(0.0346)
exit_secundaria × dist_bin = 0_100	0.0146	0.0146	0.0146	0.0146
	(0.0513)	(0.0537)	(0.0481)	(0.0567)
exit_secundaria × dist_bin = 100_200	0.0480	0.0480	0.0480	0.0480
	(0.0461)	(0.0457)	(0.0448)	(0.0466)
exit_secundaria × dist_bin = 200_300	0.0697	0.0697	0.0697	0.0697
	(0.0495)	(0.0487)	(0.0505)	(0.0509)
start × dist_bin = 0_100	-0.2382***	-0.2382***	-0.2382***	-0.2382***
	(0.0759)	(0.0725)	(0.0745)	(0.0664)
start × dist_bin = 100_200	-0.2553***	-0.2553***	-0.2553***	-0.2553***
	(0.0715)	(0.0718)	(0.0753)	(0.0770)
start × dist_bin = 200_300	-0.1993**	-0.1993**	-0.1993***	-0.1993***
	(0.0792)	(0.0779)	(0.0731)	(0.0677)
Standard-Errors	grid_id	grid_id & date	0.5km	1km
Observations	1,174,128	1,174,128	1,174,128	1,174,128
Log-Likelihood	-220,987.7	-220,987.7	-220,987.7	-220,987.7
grid_id fixed effects	✓	✓	✓	✓
date fixed effects	✓	✓	✓	✓

Notes: Poisson QMLE with grid and date fixed effects. Dependent variable: robbery count per grid-day-slot. Col. (1) cluster SE by grid; col. (2) two-way cluster grid × date; col. (3) Conley HAC 0.5 km; col. (4) Conley HAC 1 km. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 7: Spatial gradient (`z_dist2`): alternative standard errors

	n_crimes			
	Cluster grid	Twoway	Conley 0.5km	Conley 1km
	(1)	(2)	(3)	(4)
exit_primaria	-1.420***	-1.420***	-1.420***	-1.420***
	(0.1015)	(0.1015)	(0.0959)	(0.0850)
exit_secundaria	-2.066***	-2.066***	-2.066***	-2.066***
	(0.1314)	(0.1314)	(0.1284)	(0.1470)
start	-2.105***	-2.105***	-2.105***	-2.105***
	(0.1529)	(0.1546)	(0.1466)	(0.1655)
exit_primaria × dist_bin = 0_100	0.1253***	0.1253***	0.1253***	0.1253***
	(0.0420)	(0.0424)	(0.0425)	(0.0404)
exit_primaria × dist_bin = 100_200	0.1330***	0.1330***	0.1330***	0.1330***
	(0.0401)	(0.0417)	(0.0430)	(0.0386)
exit_primaria × dist_bin = 200_300	0.1011**	0.1011**	0.1011**	0.1011***
	(0.0418)	(0.0416)	(0.0403)	(0.0357)
exit_secundaria × dist_bin = 0_100	0.0218	0.0218	0.0218	0.0218
	(0.0527)	(0.0552)	(0.0511)	(0.0617)
exit_secundaria × dist_bin = 100_200	0.0516	0.0516	0.0516	0.0516
	(0.0471)	(0.0467)	(0.0456)	(0.0472)
exit_secundaria × dist_bin = 200_300	0.0768	0.0768	0.0768	0.0768
	(0.0500)	(0.0498)	(0.0510)	(0.0522)
start × dist_bin = 0_100	-0.2001***	-0.2001***	-0.2001***	-0.2001***
	(0.0745)	(0.0719)	(0.0746)	(0.0663)
start × dist_bin = 100_200	-0.2187***	-0.2187***	-0.2187***	-0.2187***
	(0.0727)	(0.0734)	(0.0746)	(0.0705)
start × dist_bin = 200_300	-0.1922**	-0.1922**	-0.1922***	-0.1922***
	(0.0793)	(0.0782)	(0.0718)	(0.0668)
exit_primaria × dist_police_bin = 0_100	0.0655	0.0655	0.0655	0.0655
	(0.0683)	(0.0670)	(0.0697)	(0.0676)
exit_primaria × dist_police_bin = 100_200	0.1061	0.1061	0.1061	0.1061
	(0.0833)	(0.0834)	(0.0814)	(0.0717)
exit_primaria × dist_police_bin = 200_300	0.0398	0.0398	0.0398	0.0398
	(0.0350)	(0.0359)	(0.0326)	(0.0404)
exit_secundaria × dist_police_bin = 0_100	0.0983	0.0983	0.0983	0.0983
	(0.0718)	(0.0710)	(0.0703)	(0.0677)
exit_secundaria × dist_police_bin = 100_200	0.0589	0.0589	0.0589	0.0589
	(0.0762)	(0.0773)	(0.0678)	(0.0712)
exit_secundaria × dist_police_bin = 200_300	-0.0522	-0.0522	-0.0522	-0.0522
	(0.0567)	(0.0583)	(0.0602)	(0.0624)
start × dist_police_bin = 0_100	-0.2527*	-0.2527**	-0.2527**	-0.2527**
	(0.1295)	(0.1271)	(0.1285)	(0.1198)
start × dist_police_bin = 100_200	-0.0739	-0.0739	-0.0739	-0.0739
	(0.0963)	(0.0959)	(0.0769)	(0.1015)
start × dist_police_bin = 200_300	-0.2019	-0.2019	-0.2019	-0.2019
	(0.1271)	(0.1261)	(0.1449)	(0.1807)
Standard-Errors	grid_id	grid_id & date	0.5km	1km
Observations	1,174,128	1,174,128	1,174,128	1,174,128
Log-Likelihood	-220,957.9	-220,957.9	-220,957.9	-220,957.9
grid_id fixed effects	✓	✓	✓	✓
date fixed effects	✓	✓	✓	✓

Notes: See Table 6.

A.2 Robustness to Clustering Strategy and Outcome

Table 8 replicates the baseline spatial gradient (`z_dist1`) under alternative clustering strategies and outcome variables. Columns compare grid-level clustering against barrio-level clustering, and *n_crimes* against *n_hurtos*. The spatial gradient is stable across all four columns.

Table 8: Spatial gradient: robustness to clustering and outcome

Dependent Variables:	n_crimes		n_robos	n_hurtos
Model:	(1)	(2)	(3)	(4)
Variables				
franja_4cat = exit_primaria × dist_bin = 0_100	0.106**	0.106**	0.095*	0.161*
	(0.047)	(0.042)	(0.055)	(0.085)
franja_4cat = exit_primaria × dist_bin = 100_200	0.127***	0.127**	0.070	0.211**
	(0.044)	(0.051)	(0.062)	(0.085)
franja_4cat = exit_primaria × dist_bin = 200_300	0.111**	0.111**	0.160**	0.090
	(0.050)	(0.046)	(0.066)	(0.095)
franja_4cat = exit_secundaria × dist_bin = 0_100	0.031	0.031	-0.064	0.123
	(0.062)	(0.057)	(0.082)	(0.124)
franja_4cat = exit_secundaria × dist_bin = 100_200	0.066	0.066	0.025	0.133
	(0.059)	(0.052)	(0.078)	(0.121)
franja_4cat = exit_secundaria × dist_bin = 200_300	0.121**	0.121**	0.097	0.079
	(0.061)	(0.056)	(0.084)	(0.129)
franja_4cat = start × dist_bin = 0_100	-0.248***	-0.248***	-0.275***	-0.227
	(0.086)	(0.088)	(0.089)	(0.150)
franja_4cat = start × dist_bin = 100_200	-0.246***	-0.246***	-0.196**	-0.251*
	(0.081)	(0.088)	(0.078)	(0.148)
franja_4cat = start × dist_bin = 200_300	-0.194**	-0.194***	-0.236***	-0.100
	(0.091)	(0.071)	(0.078)	(0.158)
Fixed-effects				
grid_id	Yes	Yes	Yes	Yes
date	Yes	Yes	Yes	Yes
Fit statistics				
Observations	665,916	665,916	651,568	629,624
Squared Correlation	0.16297	0.16297	0.07926	0.13704
Pseudo R^2	0.21353	0.21353	0.16782	0.23418
BIC	279,593.7	279,593.7	167,178.1	114,660.6

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

A.3 Grid-Size Robustness: 500 m vs. 300 m Partition

Figure 11 overlays coefficient plots from the baseline 500 m grid ($N = 907$ cells) and the 300 m robustness grid ($N = 1{,}847$ cells) for the franja × distance-bin interaction terms. Finer spatial aggregation increases precision without altering the direction or relative ordering of estimates, supporting the conclusion that results are not an artefact of grid choice.

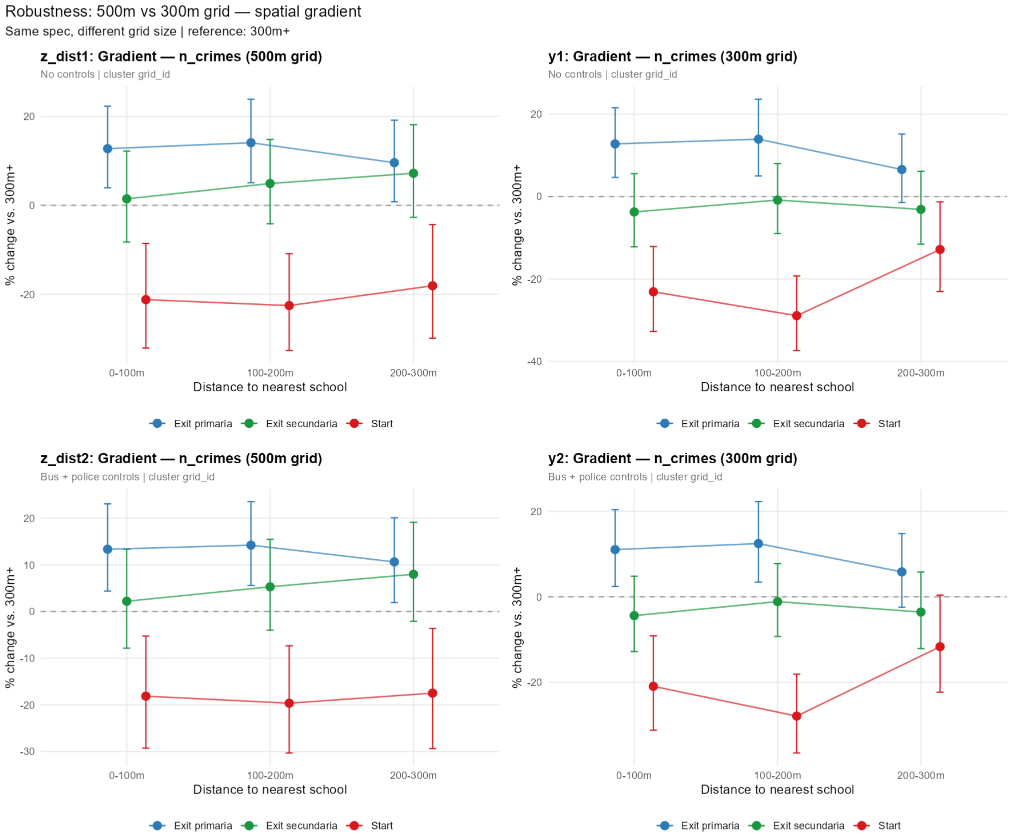

Figure 11: Coefficient comparison: 500 m grid vs. 300 m grid. Each panel shows estimates for one franja; error bars are 95 % CI clustered by grid cell.

A.4 Border-Grid Robustness

Grid cells intersecting the administrative perimeter of CABA are geometrically truncated: their effective area is smaller than 500×500 m, which mechanically reduces crime counts and artificially lowers school-density variables relative to interior cells. Moreover, buffer-based exposure measures computed for border cells may capture schools located in the *Área Metropolitana de Buenos Aires* (AMBA), outside the Mapa del Delito coverage area, inflating $N_g^{\mathrm{buf},r}$ without a corresponding increase in guardianship intensity. To assess whether this edge artefact drives the main results, I replicate the baseline specifications on the subsample of strictly interior cells—those whose full geometry lies within the CABA boundary (N = interior cells).[3]

Tables 9 and 10 present the distance-bin and exposure specifications respectively, each with two columns: (1) the full 907-cell sample with original buffer counts; (2) the interior subsample retaining original buffer counts; and (3) the interior subsample with buffer counts recomputed after excluding schools located in border cells. Point estimates are stable across all three columns in sign, magnitude, and statistical significance. The main spatial gradient—monotone decay of crime counts with distance from the nearest school during school-hour transition windows—is therefore not an artefact of geometrically trun-

[3] Border cells are identified via `st_within()`: a cell is classified as interior only if its entire polygon is contained in `st_union(barrios_sf)`. Schools located in border cells are also excluded from the buffer count recomputed for this robustness sample, so that $N_g^{\mathrm{buf},r}$ in the interior sample reflects only schools whose centroid falls within an interior cell.

cated border cells or of AMBA schools inflating buffer exposure without a corresponding increase in guardianship intensity.

Table 9: Border-grid robustness: distance-bin specifications

Dependent Variable:	n_crimes	
	Baseline bins	Control bins
Model:	(1)	(2)
Variables		
exit_primaria	-1.329***	-1.342***
	(0.0401)	(0.1272)
exit_secundaria	-1.979***	-2.076***
	(0.0520)	(0.1319)
start	-2.215***	-2.291***
	(0.0681)	(0.2232)
exit_primaria × 0_100	0.1209**	0.1217**
	(0.0484)	(0.0487)
exit_primaria × 100_200	0.1404***	0.1422***
	(0.0455)	(0.0454)
exit_primaria × 200_300	0.1307***	0.1392***
	(0.0506)	(0.0505)
exit_secundaria × 0_100	0.0331	0.0466
	(0.0651)	(0.0660)
exit_secundaria × 100_200	0.0673	0.0773
	(0.0621)	(0.0617)
exit_secundaria × 200_300	0.1388**	0.1548**
	(0.0641)	(0.0647)
start × 0_100	-0.2659***	-0.2407***
	(0.0883)	(0.0865)
start × 100_200	-0.2568***	-0.2302***
	(0.0842)	(0.0829)
start × 200_300	-0.2173**	-0.2220**
	(0.0945)	(0.0925)
Fixed-effects		
grid_id	Yes	Yes
date	Yes	Yes
Fit statistics		
Observations	608,524	608,524
Squared Correlation	0.16388	0.16401
Pseudo R^2	0.20859	0.20869
BIC	268,427.9	268,632.9

Notes: Poisson QMLE, grid and date FE. Full 907-cell sample vs. interior cells. Cluster SE by grid. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 10: Border-grid robustness: exposure specifications

Dependent Variables:	n_crimes	n_robos
	Exposure baseline	Exposure controls
Model:	(1)	(2)
Variables		
exposure_100m × exit_primaria	-0.0770	0.2546
	(0.2919)	(0.2699)
exposure_100m × exit_secundaria	-0.1466	0.2552
	(0.2768)	(0.4488)
exposure_100m × start	0.0299	-0.0148
	(0.3557)	(0.4579)
exposure_200m × exit_primaria	0.2906	-0.2907
	(0.2672)	(0.2650)
exposure_200m × exit_secundaria	0.2110	-0.1303
	(0.2663)	(0.3959)
exposure_200m × start	-0.3416	-0.1338
	(0.3363)	(0.4229)
exposure_300m × exit_primaria	-0.1283	0.2988
	(0.1588)	(0.2225)
exposure_300m × exit_secundaria	-0.1695	0.0163
	(0.1790)	(0.2851)
exposure_300m × start	-0.1320	-0.2792
	(0.2262)	(0.3104)
exit_primaria	-1.263***	-1.430***
	(0.0581)	(0.1870)
exit_secundaria	-1.844***	-2.221***
	(0.0628)	(0.2968)
start	-2.085***	-2.137***
	(0.0952)	(0.3292)
Fixed-effects		
grid_id	Yes	Yes
date	Yes	Yes
Fit statistics		
Observations	944,000	600,084
Squared Correlation	0.15366	0.07970
Pseudo R^2	0.21223	0.16426
BIC	374,746.0	159,872.9

Clustered (grid_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*
Notes: See Table 9.

Notes: Poisson QMLE, grid and date FE. Full 907-cell sample vs. interior cells. Distance-bin interactions for nearest bus stop and police station included in col. (2) but omitted from display.

A.5 Alternative School-Hour Definition

The main specification defines school-hour windows as entry (07:00), midday dismissal (12:00–13:00), and afternoon dismissal (16:00), with control hours drawn from mid-morning slots (06:00, 08:00–11:00, 14:00–15:00, 17:00–18:00). To assess sensitivity to this boundary choice, Table 11 replicates the baseline specifications under a wider definition that extends the entry window to include 08:00 and the midday window to include 13:00, with the control group adjusted accordingly. Coefficients are stable in sign, magnitude, and significance across both definitions, confirming that the main results are not driven by the precise delineation of transition windows.

Table 11: Robustness: alternative school-hour definition

Dependent Variables:	n_robos		n_crimes_v2
	Main spec (v1)	Main spec (v1) robos	wider windows (v2)
Model:	(1)	(2)	(3)
Variables			
exit_primaria	-1.356***	-1.495***	
	(0.0476)	(0.1547)	
exit_secundaria	-1.843***	-1.923***	
	(0.0556)	(0.2131)	
start	-2.040***	-2.189***	
	(0.0762)	(0.2453)	
exit_primaria × dist_bin = 0_100	0.1068*	0.0956	
	(0.0568)	(0.0583)	
exit_primaria × dist_bin = 100_200	0.0883	0.0828	
	(0.0547)	(0.0555)	
exit_primaria × dist_bin = 200_300	0.0352	0.0280	
	(0.0592)	(0.0596)	
exit_secundaria × dist_bin = 0_100	-0.0951	-0.1089	
	(0.0699)	(0.0700)	
exit_secundaria × dist_bin = 100_200	-0.0229	-0.0350	
	(0.0658)	(0.0656)	
exit_secundaria × dist_bin = 200_300	-0.0516	-0.0640	
	(0.0712)	(0.0715)	
start × dist_bin = 0_100	-0.1910**	-0.1887**	
	(0.0935)	(0.0945)	
start × dist_bin = 100_200	-0.2923***	-0.2988***	
	(0.0895)	(0.0888)	
start × dist_bin = 200_300	-0.1314	-0.1281	
	(0.0937)	(0.0936)	
is_school_time_v2			-0.9870***
			(0.0175)
is_school_time_v2 × n_schools			0.0024
			(0.0067)
is_school_time_v2 × n_bus_stops			-0.0005
			(0.0028)
is_school_time_v2 × n_police			0.0019
			(0.0263)
is_school_time_v2 × n_commercial			0.0063
			(0.0072)
Fixed-effects			
grid_id	Yes	Yes	Yes
date	Yes	Yes	Yes
Fit statistics			
Observations	2,351,740	2,351,740	2,330,500
Squared Correlation	0.03594	0.03595	0.03744
Pseudo R^2	0.14428	0.14432	0.10926
BIC	286,017.7	286,269.5	440,239.4

Clustered (grid_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Poisson QMLE, grid and date FE. Cluster SE by grid. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

B Additional Specifications

B.1 Four-Category Franja Specifications

The body of the paper uses a three-franja coding (start / exit_primaria / exit_secundaria). Table 12 presents three specifications using the four-category franja coding (start / exit_primaria / exit_secundaria / control). Column (1) (`m_A5`) includes franja interactions with aggregate school count; column (2) (`m_A6`) disaggregates by sector (public vs. private); column (3) (`m_A9`) adds triple interactions between sector, franja, and poverty rate. The main interaction patterns from the baseline distance-bin specifications are preserved across all three columns. Notably, the triple interaction `start` $\times$ N_g^{pub} $\times$ `poverty_rate` is positive and significant ($\hat{\beta} = 0.369$, $p < 0.001$), suggesting that in high-poverty grids, public school entry hours are associated with *higher* crime relative to the control window—consistent with a congestion effect that overwhelms guardianship when enrolment is large and informal supervision capacity is low.

Table 12: Four-franja specification: baseline, sector, and poverty

Dependent Variable:	n_crimes		
	Franjas	Sector	Sector + Poverty
Model:	(1)	(2)	(3)
Variables			
exit_primaria	-1.27***	-1.26***	-1.24***
	(0.027)	(0.028)	(0.028)
exit_secundaria	-1.92***	-1.93***	-1.89***
	(0.033)	(0.034)	(0.032)
start	-2.32***	-2.34***	-2.37***
	(0.049)	(0.051)	(0.048)
n_commercial × exit_primaria	0.008**	0.007*	0.015***
	(0.004)	(0.004)	(0.004)
n_commercial × exit_secundaria	-0.014**	-0.013*	-0.003
	(0.007)	(0.007)	(0.008)
n_commercial × start	-0.044***	-0.041***	-0.048***
	(0.012)	(0.013)	(0.013)
n_pub_schools × exit_primaria		-0.0004	
		(0.008)	
n_pub_schools × exit_secundaria		0.018*	
		(0.010)	
n_pub_schools × start		0.010	
		(0.017)	
n_priv_schools × exit_primaria		0.015**	
		(0.007)	
n_priv_schools × exit_secundaria		0.009	
		(0.011)	
n_priv_schools × start		-0.033**	
		(0.015)	
n_pub_schools × poverty_rate × exit_primaria			-0.004
			(0.076)
n_pub_schools × poverty_rate × exit_secundaria			0.028
			(0.082)
n_pub_schools × poverty_rate × start			0.275**
			(0.134)
Fixed-effects			
grid_id	Yes	Yes	Yes
date	Yes	Yes	Yes
Fit statistics			
Observations	665,916	665,916	662,540
Squared Correlation	0.16312	0.16309	0.16289
Pseudo R^2	0.21370	0.21371	0.21288
BIC	279,576.1	279,572.2	279,399.8

Clustered (grid_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Poisson QMLE, grid and date FE. Controls for bus stops, police stations, and commercial activity interacted with franja omitted. Cluster SE by grid. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

C Additional Figures

C.1 Heterogeneity: School Sector and Poverty

Figures 21 and 22 display coefficient plots for the public vs. private school sector split and the poverty interaction, respectively. The sector split (Figure 21) reveals that the proximity effect is concentrated around public schools, consistent with their larger average enrolment and more heterogeneous dismissal schedules. The poverty interaction (Figure 22) shows an amplified effect in higher-poverty cells, in line with Kelly (2000).

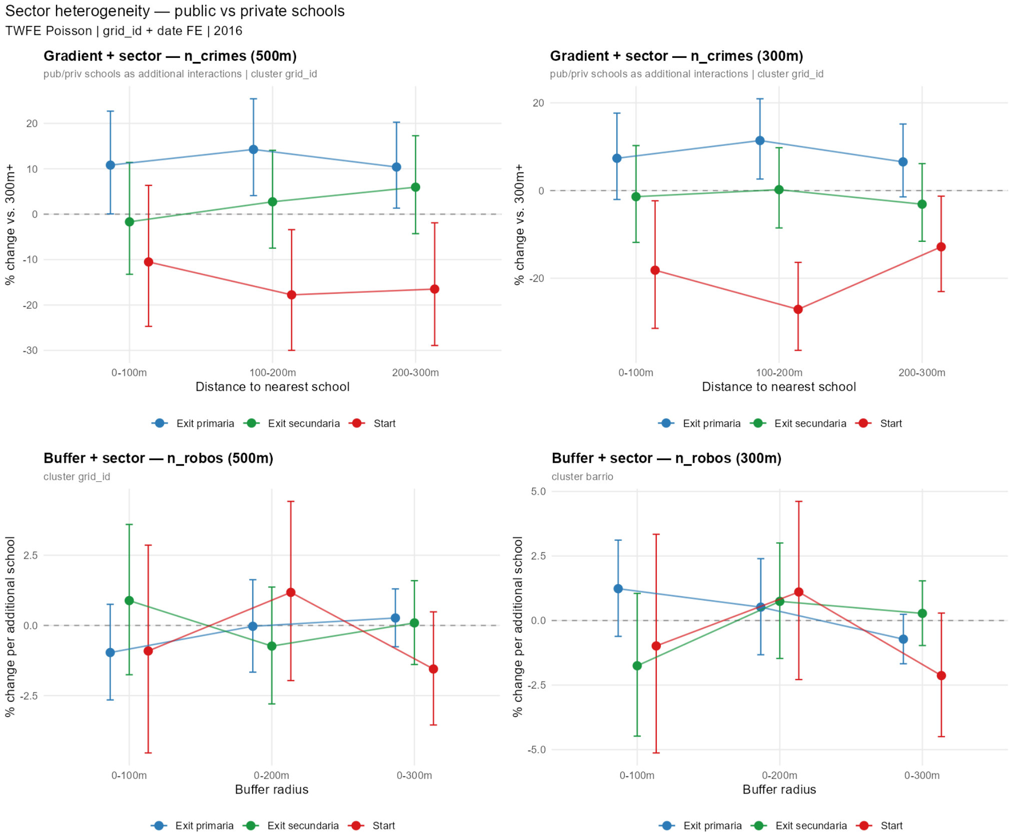

Figure 12: Heterogeneity by school sector (public vs. private). Coefficients from the franja × distance-bin model estimated separately for public and private school density. 95 % CI clustered by grid cell.

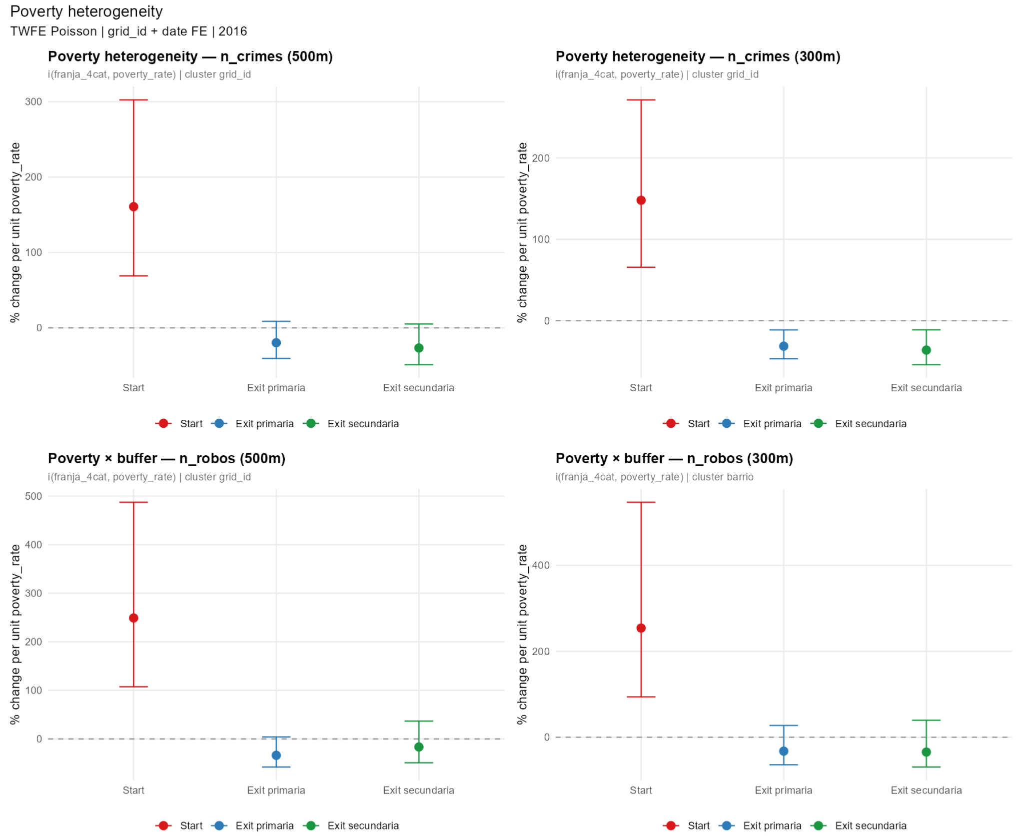

Figure 13: Heterogeneity by poverty rate. Marginal effects of the triple interaction $N_g^{\text{schools}} \times \texttt{SchoolHour}_t \times \texttt{poverty\_rate}_g$ evaluated at the 25th, 50th, and 75th percentiles of the poverty distribution. 95 % CI clustered by grid cell.

C.2 Buffer Model Comparison: 2×2 Panel

Figure 14 presents a four-panel comparison of the distance-gradient coefficient plots across buffer radio of 100 m, 150 m, 200 m, and 300 m. Monotone decay is visible across all radio. The effect attenuates beyond 200 m, consistent with a pedestrian catchment of roughly two city blocks surrounding each school.

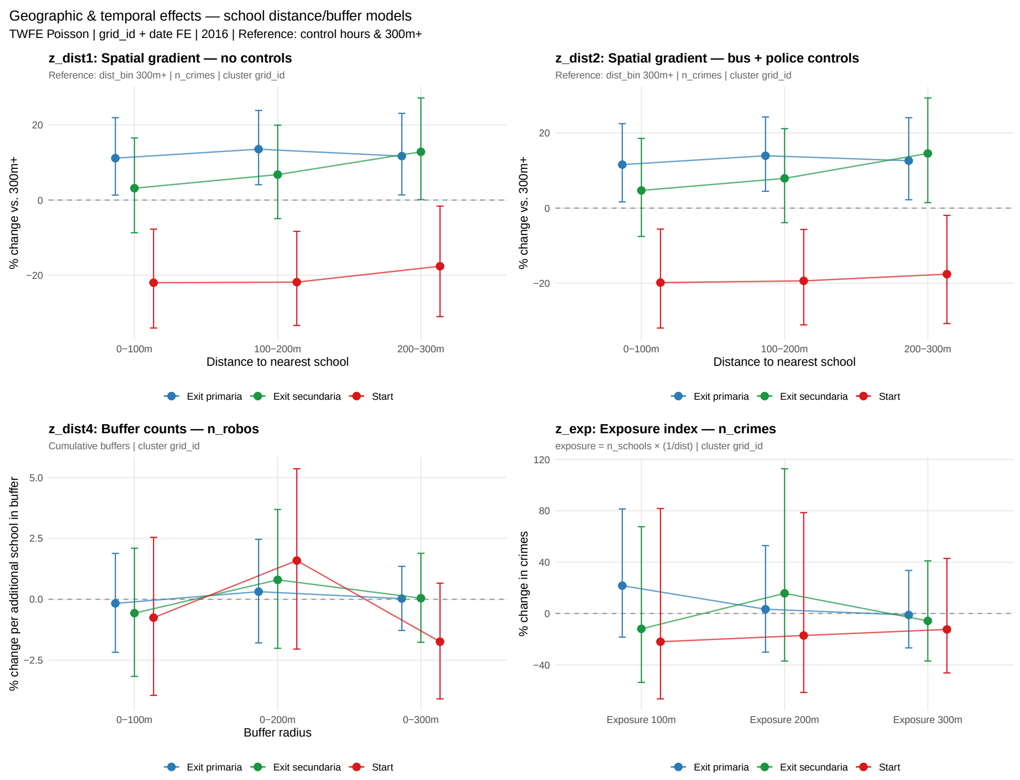

Figure 14: Buffer model comparison (2 × 2 panel). Each subplot shows the franja × distance-bin coefficients for the indicated buffer radius. 95 % CI clustered by grid cell.

D Senderos Escolares: Additional Results

D.1 Sun-Abraham ATT by Slot Category

Table 13 reports the Sun-Abraham bimonthly ATT estimates for the four slot subsets. The intensity= 0 subset (prevention agents only) yields consistently negative post-treatment coefficients, while the intensity= 1 subset (agents + transit police) is indistinguishable from zero throughout, identifying the guardianship channel as the operative mechanism.

Table 13: Sun-Abraham ATT by slot category (bimonthly panel, *n_robos*)

Model:	Intensity=0 (1)	Intensity=1 (2)	School hrs (3)	Non-school hrs (4)
Variables				
Constant	0.265***	0.108*	-0.183***	0.460***
	(0.057)	(0.058)	(0.056)	(0.058)
bim_idx = -6	-0.086	-0.012	-0.004	-0.076
	(0.067)	(0.071)	(0.068)	(0.070)
bim_idx = -5	-0.088	0.054	-0.029	-0.015
	(0.062)	(0.066)	(0.066)	(0.063)
bim_idx = -4	-0.093	0.069	-0.022	-0.011
	(0.066)	(0.068)	(0.069)	(0.066)
bim_idx = -3	-0.110*	0.0010	-0.076	-0.047
	(0.062)	(0.068)	(0.070)	(0.062)
bim_idx = -2	-0.030	0.0001	0.024	-0.038
	(0.069)	(0.068)	(0.069)	(0.067)
bim_idx = 0	-0.145**	-0.102	-0.105	-0.135**
	(0.062)	(0.064)	(0.066)	(0.062)
bim_idx = 1	-0.202***	-0.073	-0.136**	-0.142**
	(0.064)	(0.064)	(0.065)	(0.062)
bim_idx = 2	-0.183***	-0.134**	-0.217***	-0.132**
	(0.064)	(0.067)	(0.072)	(0.062)
bim_idx = 3	-0.141**	-0.080	-0.065	-0.137**
	(0.063)	(0.066)	(0.068)	(0.062)
bim_idx = 4	-0.138**	-0.098	-0.118*	-0.120*
	(0.062)	(0.067)	(0.066)	(0.062)
bim_idx = 6	-0.049	-0.025	-0.012	-0.051
	(0.061)	(0.064)	(0.065)	(0.061)
bim_idx = 7	-0.145**	0.021	-0.112	-0.041
	(0.065)	(0.068)	(0.068)	(0.064)
bim_idx = 8	0.045	0.036	0.075	0.022
	(0.064)	(0.067)	(0.067)	(0.064)
bim_idx = 9	0.048	0.050	0.102	0.021
	(0.061)	(0.064)	(0.063)	(0.062)
bim_idx = 10	0.171***	0.031	0.110*	0.107*
	(0.059)	(0.066)	(0.062)	(0.062)
bim_idx = 12	0.068	-0.092	-0.067	0.029
	(0.060)	(0.062)	(0.062)	(0.061)
bim_idx = 13	0.040	-0.064	-0.013	-0.004
	(0.060)	(0.065)	(0.062)	(0.062)
bim_idx = 14	-0.145**	-0.182***	-0.138**	-0.175***
	(0.059)	(0.066)	(0.066)	(0.060)
bim_idx = 15	-0.121*	-0.145**	-0.113*	-0.142**
	(0.063)	(0.062)	(0.064)	(0.061)
bim_idx = 16	-0.667***	-0.759***	-0.675***	-0.727***
	(0.065)	(0.069)	(0.072)	(0.063)
n_bus_stops	0.048***	0.049***	0.048***	0.048***
	(0.003)	(0.003)	(0.003)	(0.003)
Fit statistics				
Observations	20,520	20,520	20,520	20,520
Squared Correlation	0.14539	0.15760	0.14790	0.18459
Pseudo R^2	0.08674	0.08849	0.07462	0.10261
BIC	86,045.2	77,523.1	65,487.4	90,405.6

Clustered (grid_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Poisson PPML. Clustered SE by grid. Cohort = Fase Original (id $\leq$ 207), threshold 200m. Reference: B6 (Nov-Dic 2016). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

D.2 Entry vs. Exit Slot Decomposition

Figure 9 decomposes the Sun-Abraham ATT by school transition type: morning entry (7–8 h), primary exit (12 h), and secondary exit (16–17 h). The three series move in parallel with no systematic ordering, indicating that the programme effect is not concentrated in a single transition type and operates through sustained route-level guardianship throughout the school day.

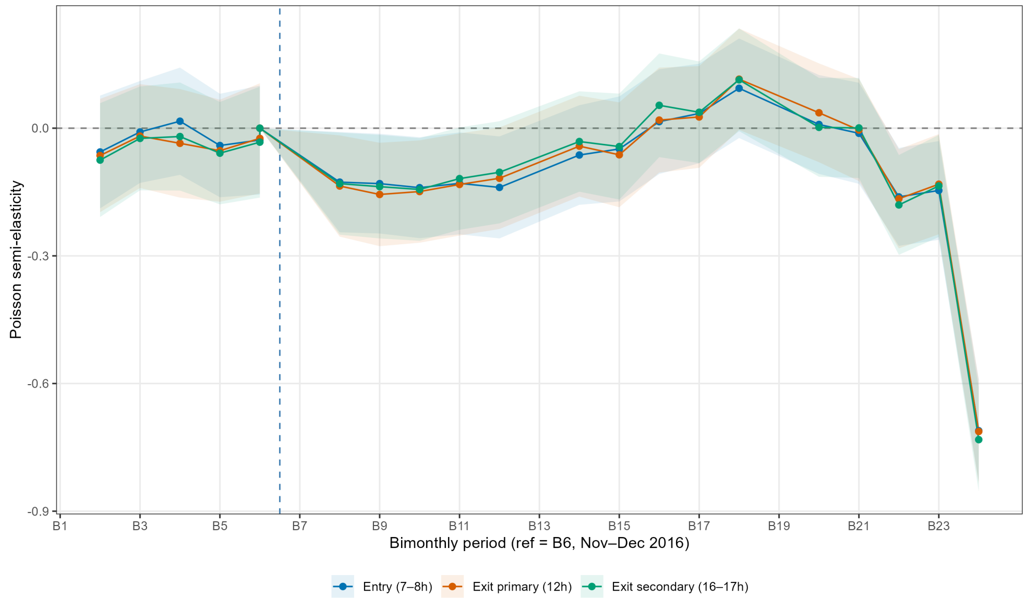

Figure 15: Sun-Abraham ATT by transition type. Entry (7–8 h), primary exit (12 h), secondary exit (16–17 h). Reference $b = 6$ (Nov–Dec 2016). 95 % CI clustered by grid. Outcome: *n_robos*.

D.3 Dose Threshold Sensitivity

Table 14 presents the post-period Wald statistics for the Sun-Abraham ATT under five alternative dose thresholds ($\ell^* \in \{50, 100, 200, 300, 400\}$ m) for the `intensity`= 0 subset. All five thresholds reject the null of zero post-treatment effects ($p < 2.2 \times 10^{-16}$ throughout). The Wald statistic declines monotonically from $F = 46.6$ at $\ell^* = 50$ m to $F = 28.9$ at $\ell^* = 400$ m, reflecting the progressive reduction in the number of treated grids as the threshold excludes cells with lower route coverage—at higher thresholds the same routes generate a smaller treated sample, reducing statistical power without changing the qualitative result. The stability across all thresholds confirms that the crime reduction is not driven by grids with minimal route presence being miscoded as treated, and that the 200 m baseline represents a conservative middle point in the threshold sensitivity range.

Table 14: SA sensitivity to dose threshold (intensity= 0, n_robos)

Model:	50m (1)	100m (2)	200m (3)	300m (4)	400m (5)
Variables					
Constant	0.162***	0.190***	0.265***	0.288***	0.275***
	(0.055)	(0.056)	(0.057)	(0.061)	(0.066)
bim_idx = -6	0.030	0.011	-0.086	-0.040	-0.077
	(0.065)	(0.067)	(0.067)	(0.072)	(0.079)
bim_idx = -5	-0.018	-0.037	-0.088	-0.055	-0.097
	(0.062)	(0.063)	(0.062)	(0.064)	(0.068)
bim_idx = -4	-0.004	-0.022	-0.093	-0.060	-0.114
	(0.064)	(0.065)	(0.066)	(0.068)	(0.075)
bim_idx = -3	-0.001	-0.021	-0.110*	-0.082	-0.142**
	(0.064)	(0.065)	(0.062)	(0.064)	(0.068)
bim_idx = -2	0.069	0.041	-0.030	-0.0007	-0.055
	(0.067)	(0.068)	(0.069)	(0.073)	(0.079)
bim_idx = 0	-0.056	-0.079	-0.145**	-0.120*	-0.156**
	(0.064)	(0.064)	(0.062)	(0.064)	(0.069)
bim_idx = 1	-0.116*	-0.140**	-0.202***	-0.174**	-0.182***
	(0.064)	(0.065)	(0.064)	(0.068)	(0.069)
bim_idx = 2	-0.098	-0.123*	-0.183***	-0.177***	-0.229***
	(0.063)	(0.063)	(0.064)	(0.065)	(0.069)
bim_idx = 3	-0.063	-0.088	-0.141**	-0.111*	-0.156**
	(0.063)	(0.064)	(0.063)	(0.065)	(0.071)
bim_idx = 4	-0.041	-0.073	-0.138**	-0.126**	-0.171**
	(0.062)	(0.063)	(0.062)	(0.064)	(0.069)
bim_idx = 6	0.020	0.004	-0.049	-0.014	-0.049
	(0.062)	(0.062)	(0.061)	(0.064)	(0.066)
bim_idx = 7	-0.034	-0.065	-0.145**	-0.101	-0.135**
	(0.063)	(0.064)	(0.065)	(0.065)	(0.069)
bim_idx = 8	0.119*	0.105*	0.045	0.049	0.011
	(0.063)	(0.064)	(0.064)	(0.066)	(0.070)
bim_idx = 9	0.142**	0.117*	0.048	0.075	0.027
	(0.061)	(0.062)	(0.061)	(0.062)	(0.065)
bim_idx = 10	0.254***	0.240***	0.171***	0.204***	0.166**
	(0.059)	(0.060)	(0.059)	(0.061)	(0.065)
bim_idx = 12	0.157**	0.143**	0.068	0.075	0.030
	(0.061)	(0.062)	(0.060)	(0.064)	(0.070)
bim_idx = 13	0.095	0.086	0.040	0.071	0.023
	(0.060)	(0.061)	(0.060)	(0.062)	(0.066)
bim_idx = 14	-0.076	-0.085	-0.145**	-0.110*	-0.153**
	(0.060)	(0.060)	(0.059)	(0.062)	(0.067)
bim_idx = 15	-0.042	-0.066	-0.121*	-0.107	-0.181**
	(0.062)	(0.063)	(0.063)	(0.067)	(0.071)
bim_idx = 16	-0.571***	-0.596***	-0.667***	-0.619***	-0.662***
	(0.065)	(0.066)	(0.065)	(0.069)	(0.073)
n_bus_stops	0.049***	0.049***	0.048***	0.045***	0.049***
	(0.003)	(0.003)	(0.003)	(0.004)	(0.004)
Fit statistics					
Observations	24,160	22,800	20,520	17,920	15,040
Squared Correlation	0.14213	0.14534	0.14539	0.11407	0.12508
Pseudo R^2	0.08682	0.08809	0.08674	0.07182	0.07444
BIC	99,176.6	94,313.6	86,045.2	75,461.8	63,187.4

Clustered (grid_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Each column re-assigns cohorts using the indicated threshold. Poisson QMLE, Sun-Abraham estimator, grid and bimester FE. Cluster SE by grid. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

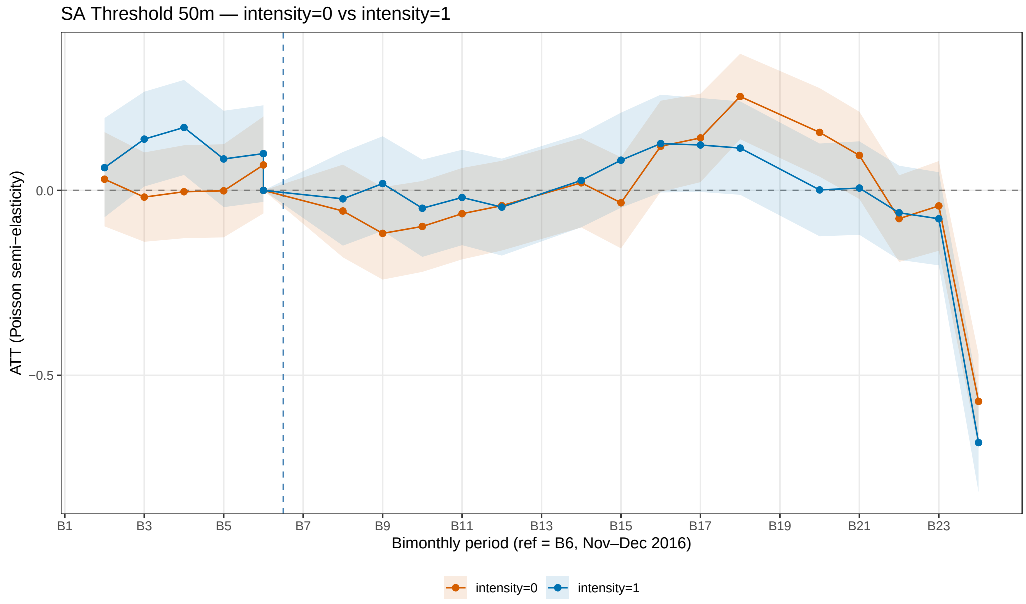

Figure 16: SA threshold $\ell^* = 50\,\mathrm{m}$. Orange: intensity$=0$; blue: intensity$=1$. Reference $b = 6$. 95 % CI clustered by grid.

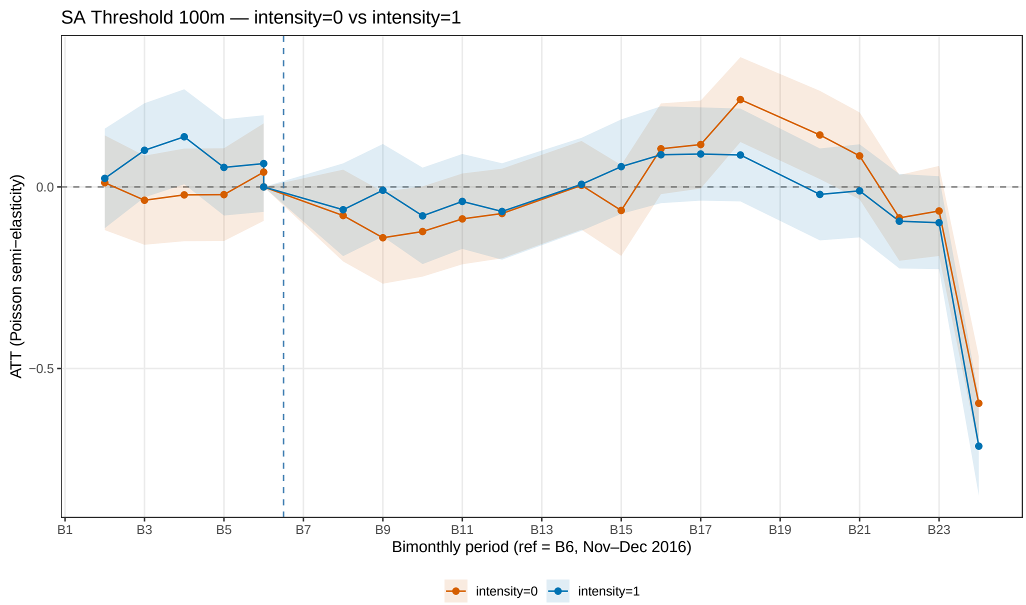

Figure 17: SA threshold $\ell^* = 100\,\mathrm{m}$. See Figure 16.

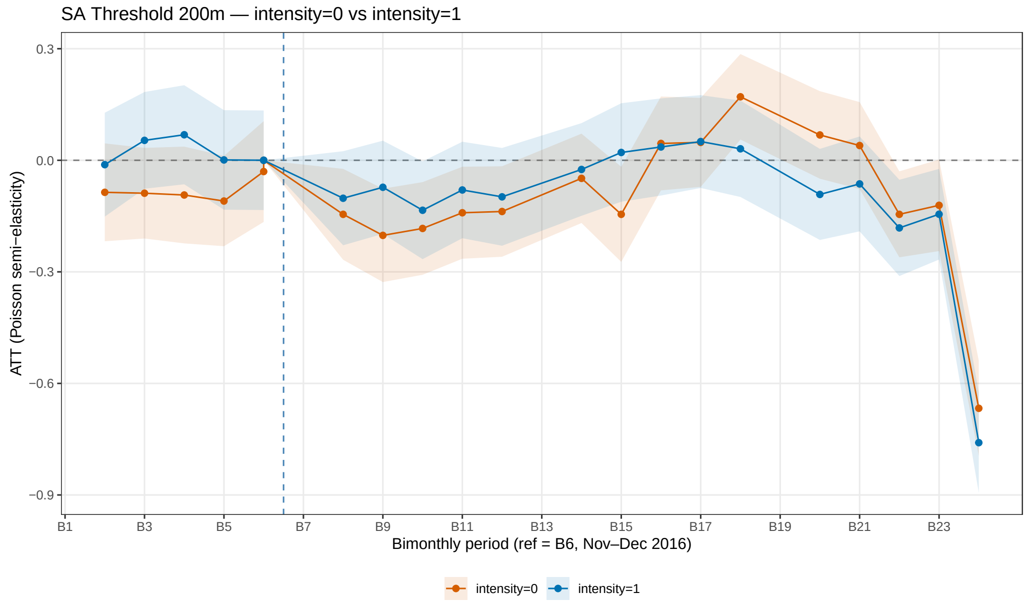

Figure 18: SA threshold $\ell^* = 200\,\text{m}$ — baseline specification. See Figure 16.

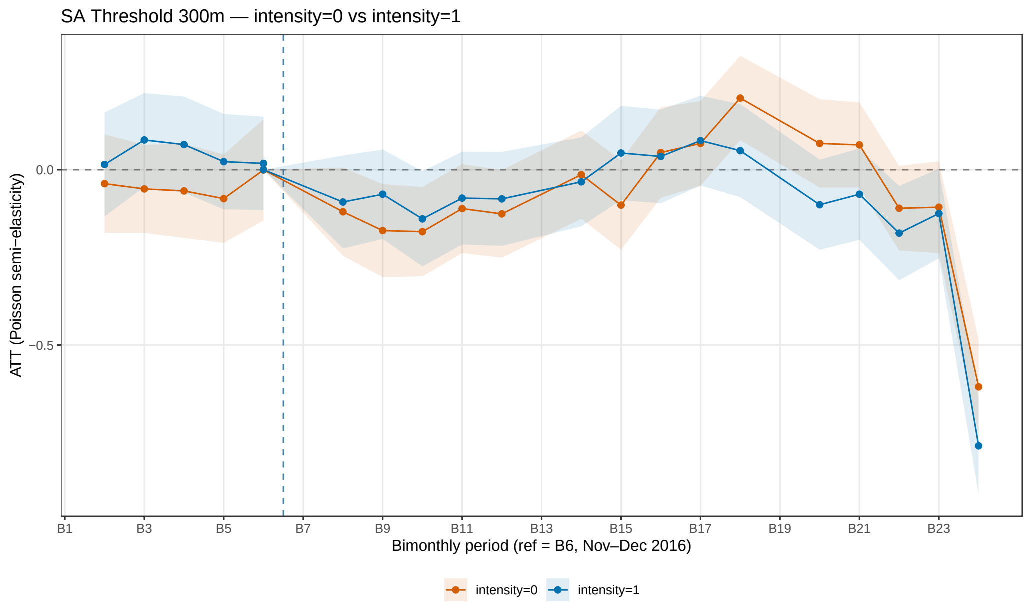

Figure 19: SA threshold $\ell^* = 300\,\text{m}$. See Figure 16.

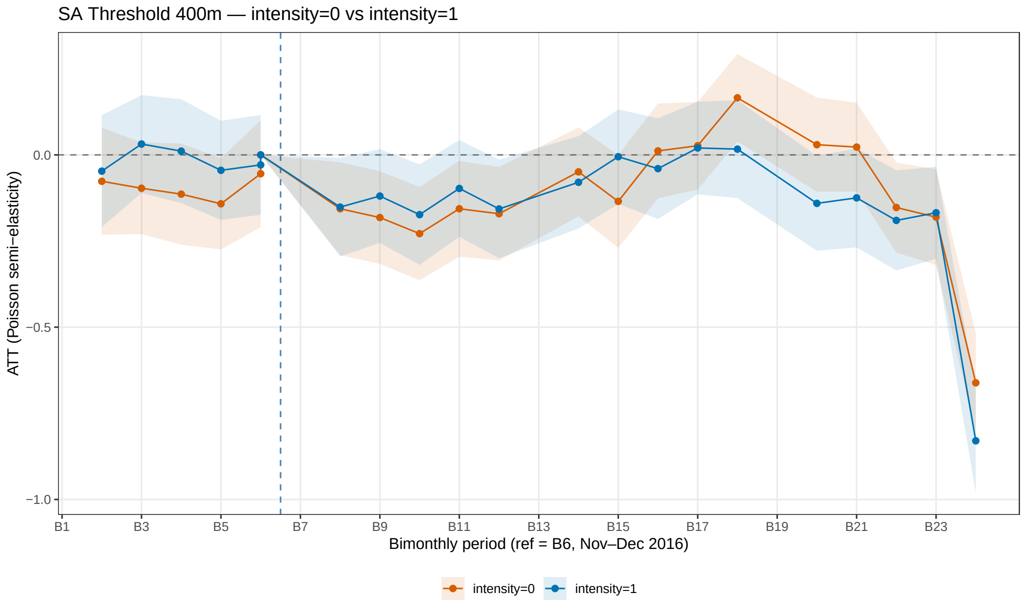

Figure 20: SA threshold $\ell^* = 400\,\mathrm{m}$. See Figure 16.

D.4 Heterogeneity by School Sector

Figure 21 overlays Sun-Abraham ATT estimates from public-dominant and private-dominant grid subgroups. Both subgroups yield negative point estimates consistent with the baseline result; standard errors are inflated by the reduction in sample size within each subgroup and no subgroup ATT achieves conventional significance thresholds. Table 15 reports the corresponding coefficient tables.

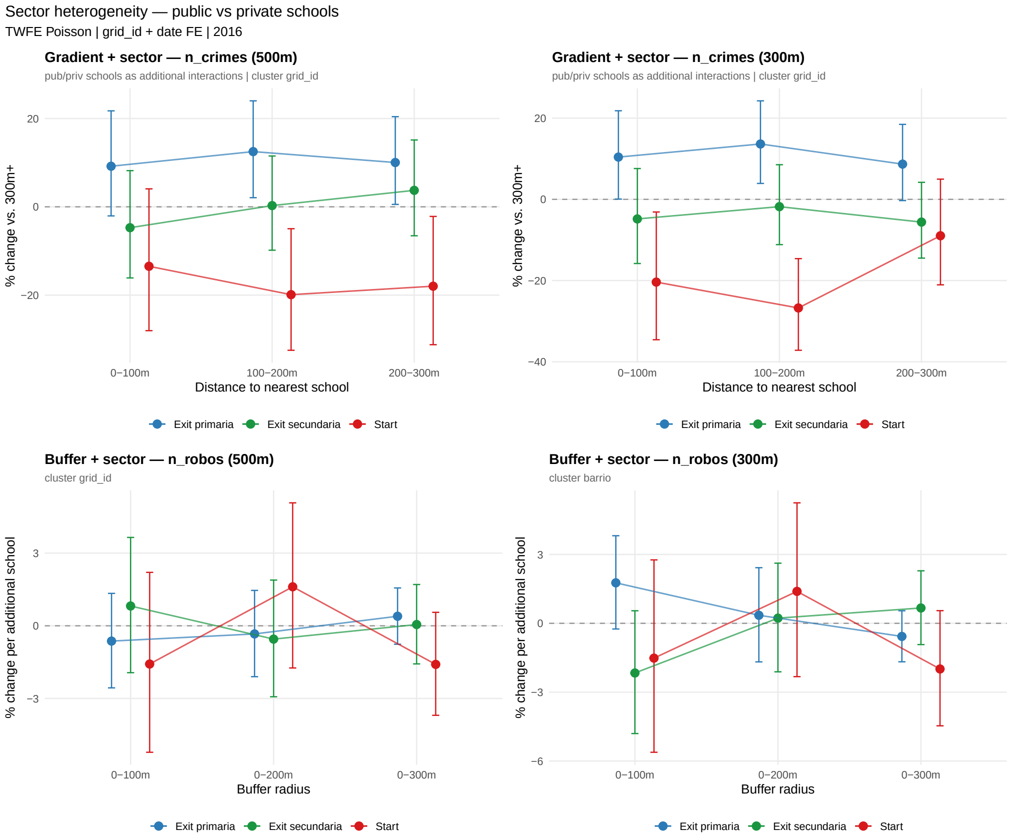

Figure 21: Heterogeneity by school sector. Public-dominant grids: $N_g^{\text{pub}} > N_g^{\text{priv}}$; private-dominant: otherwise. Sun-Abraham ATT, intensity$=1$ panel. 95 % CI clustered by grid.

Table 15: Heterogeneity by school sector (SA subgroup models)

Model:	Baseline (1)	Public-dominant (2)	Private-dominant (3)
Variables			
Constant	0.108*	0.104	0.151
	(0.058)	(0.067)	(0.110)
bim_idx = -6	-0.012	0.005	-0.112
	(0.071)	(0.098)	(0.120)
bim_idx = -5	0.054	0.017	0.093
	(0.066)	(0.093)	(0.116)
bim_idx = -4	0.069	0.038	0.117
	(0.068)	(0.093)	(0.120)
bim_idx = -3	0.0010	0.009	0.028
	(0.068)	(0.092)	(0.123)
bim_idx = -2	0.0001	0.022	-0.052
	(0.068)	(0.098)	(0.116)
bim_idx = 0	-0.102	-0.105	-0.072
	(0.064)	(0.087)	(0.116)
bim_idx = 1	-0.073	-0.092	-0.041
	(0.064)	(0.088)	(0.112)
bim_idx = 2	-0.134**	-0.126	-0.192
	(0.067)	(0.092)	(0.120)
bim_idx = 3	-0.080	-0.128	-0.044
	(0.066)	(0.088)	(0.118)
bim_idx = 4	-0.098	-0.145	-0.052
	(0.067)	(0.092)	(0.121)
bim_idx = 6	-0.025	-0.069	0.048
	(0.064)	(0.083)	(0.117)
bim_idx = 7	0.021	-0.032	0.124
	(0.068)	(0.092)	(0.116)
bim_idx = 8	0.036	0.0003	0.095
	(0.067)	(0.097)	(0.113)
bim_idx = 9	0.050	-0.042	0.165
	(0.064)	(0.085)	(0.114)
bim_idx = 10	0.031	-0.032	0.102
	(0.066)	(0.094)	(0.116)
bim_idx = 12	-0.092	-0.112	-0.112
	(0.062)	(0.087)	(0.109)
bim_idx = 13	-0.064	-0.137	-0.003
	(0.065)	(0.084)	(0.122)
bim_idx = 14	-0.182***	-0.182**	-0.136
	(0.066)	(0.089)	(0.120)
bim_idx = 15	-0.145**	-0.198**	-0.077
	(0.062)	(0.083)	(0.113)
bim_idx = 16	-0.759***	-0.794***	-0.704***
	(0.069)	(0.097)	(0.121)
n_bus_stops	0.049***	0.046***	0.051***
	(0.003)	(0.004)	(0.004)
Fit statistics			
Observations	20,520	10,760	6,000
Squared Correlation	0.15760	0.16802	0.17208
Pseudo R^2	0.08849	0.08656	0.10276
BIC	77,523.1	39,729.2	24,296.1

Clustered (grid_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Poisson QMLE. Sun-Abraham estimator. Subgroups defined by $N_g^{\text{pub}} > N_g^{\text{priv}}$ (public-dominant) and vice versa. Grid and bimester FE. Cluster SE by grid. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

D.5 Heterogeneity by Poverty Rate

Figure 22 overlays Sun-Abraham ATT estimates from high- and low-poverty grid subgroups, split at the median NBI rate among non-zero grids. As with the sector split, point estimates are negative in both subgroups but statistically uninformative due to reduced sample size. Table 16 reports coefficients.

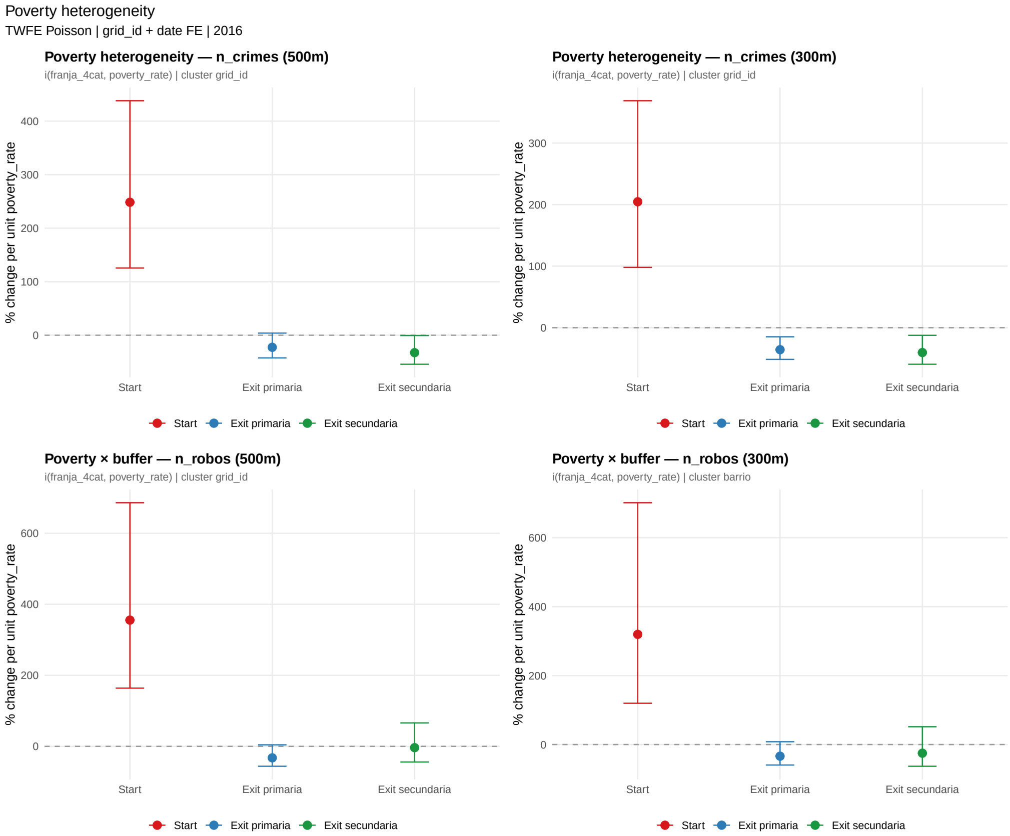

Figure 22: Heterogeneity by poverty rate. High-poverty: NBI rate at or above median among non-zero grids; low-poverty: below median. Sun-Abraham ATT, intensity= 1 panel. 95 % CI clustered by grid.

Table 16: Heterogeneity by poverty rate (SA subgroup models)

Model:	Baseline (1)	High poverty (2)	Low poverty (3)
Variables			
Constant	0.108*	0.274***	-0.064
	(0.058)	(0.073)	(0.090)
bim_idx = -6	-0.012	0.054	-0.055
	(0.071)	(0.101)	(0.085)
bim_idx = -5	0.054	0.095	0.042
	(0.066)	(0.092)	(0.083)
bim_idx = -4	0.069	0.138	0.022
	(0.068)	(0.090)	(0.090)
bim_idx = -3	0.0010	0.056	-0.028
	(0.068)	(0.092)	(0.090)
bim_idx = -2	0.0001	0.016	0.020
	(0.068)	(0.092)	(0.089)
bim_idx = 0	-0.102	-0.038	-0.143*
	(0.064)	(0.087)	(0.083)
bim_idx = 1	-0.073	-0.022	-0.096
	(0.064)	(0.087)	(0.083)
bim_idx = 2	-0.134**	-0.093	-0.145*
	(0.067)	(0.090)	(0.087)
bim_idx = 3	-0.080	-0.070	-0.053
	(0.066)	(0.093)	(0.082)
bim_idx = 4	-0.098	-0.040	-0.132
	(0.067)	(0.088)	(0.091)
bim_idx = 6	-0.025	0.063	-0.098
	(0.064)	(0.082)	(0.087)
bim_idx = 7	0.021	0.139	-0.096
	(0.068)	(0.089)	(0.086)
bim_idx = 8	0.036	0.110	-0.018
	(0.067)	(0.092)	(0.084)
bim_idx = 9	0.050	0.039	0.102
	(0.064)	(0.084)	(0.086)
bim_idx = 10	0.031	0.041	0.058
	(0.066)	(0.089)	(0.087)
bim_idx = 12	-0.092	0.009	-0.183**
	(0.062)	(0.083)	(0.083)
bim_idx = 13	-0.064	-0.044	-0.047
	(0.065)	(0.084)	(0.091)
bim_idx = 14	-0.182***	-0.173*	-0.154*
	(0.066)	(0.091)	(0.085)
bim_idx = 15	-0.145**	-0.115	-0.143*
	(0.062)	(0.086)	(0.079)
bim_idx = 16	-0.759***	-0.705***	-0.787***
	(0.069)	(0.095)	(0.090)
n_bus_stops	0.049***	0.043***	0.053***
	(0.003)	(0.003)	(0.007)
Fit statistics			
Observations	20,520	10,280	10,240
Squared Correlation	0.15760	0.14620	0.14673
Pseudo R^2	0.08849	0.07795	0.08284
BIC	77,523.1	42,874.1	34,139.5

Clustered (grid_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: See Table 15.

D.6 Crime Displacement Analysis

Table 17 and Figure 23 report the displacement effect comparing adjacent never-treated grids against non-adjacent never-treated grids. Post-treatment coefficients are mostly indistinguishable from zero in both intensity subsets, indicating weak evidence of systematic crime displacement to the immediate neighbourhood of treated routes.

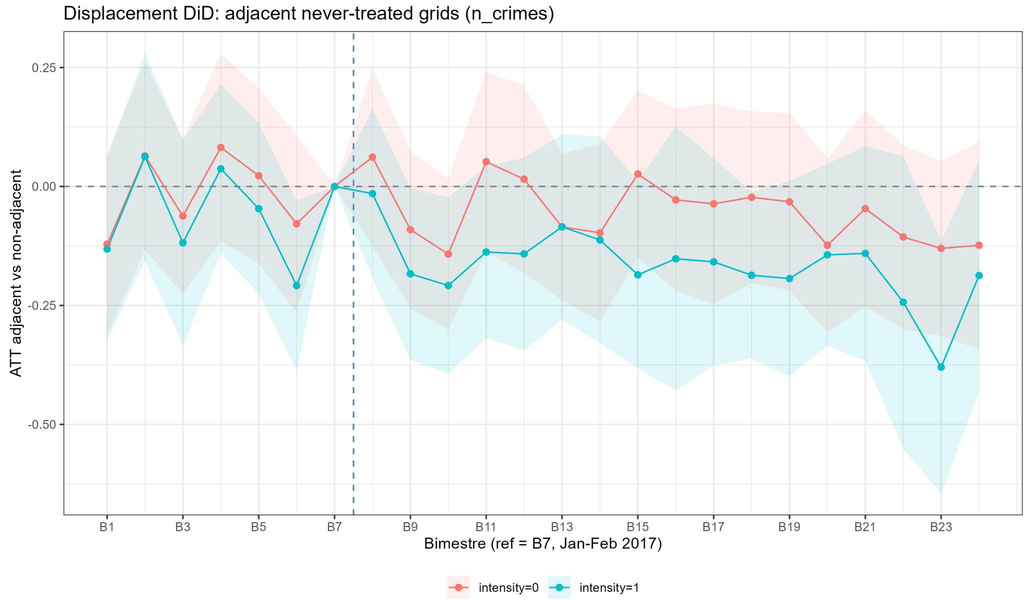

Figure 23: Displacement: adjacent vs. non-adjacent never-treated grids. Red: intensity= 0; blue: intensity= 1. Reference $b = 6$. 95 % CI clustered by grid. Outcome: n_crimes.

Table 17: Displacement: adjacent never-treated grids

Model:	Intensity=0 (1)	Intensity=1 (2)
Variables		
is_adjacent × bim_idx = 2	0.364	0.154
	(0.229)	(0.211)
is_adjacent × bim_idx = 3	-0.038	-0.039
	(0.290)	(0.203)
is_adjacent × bim_idx = 4	0.127	0.161
	(0.214)	(0.191)
is_adjacent × bim_idx = 5	0.188	0.080
	(0.222)	(0.214)
is_adjacent × bim_idx = 8	-0.350*	-0.133
	(0.199)	(0.235)
is_adjacent × bim_idx = 9	0.104	0.059
	(0.254)	(0.260)
is_adjacent × bim_idx = 10	-0.299	0.030
	(0.263)	(0.207)
is_adjacent × bim_idx = 11	-0.031	0.050
	(0.246)	(0.233)
is_adjacent × bim_idx = 12	0.086	-0.047
	(0.284)	(0.150)
is_adjacent × bim_idx = 14	-0.214	0.013
	(0.219)	(0.287)
is_adjacent × bim_idx = 15	0.457*	-0.329*
	(0.253)	(0.195)
is_adjacent × bim_idx = 16	-0.290	0.311
	(0.276)	(0.287)
is_adjacent × bim_idx = 17	0.065	-0.569***
	(0.213)	(0.220)
is_adjacent × bim_idx = 18	-0.326	-0.230
	(0.214)	(0.214)
is_adjacent × bim_idx = 20	-0.080	0.021
	(0.189)	(0.221)
is_adjacent × bim_idx = 21	-0.194	-0.261
	(0.233)	(0.237)
is_adjacent × bim_idx = 22	-0.515**	-0.163
	(0.233)	(0.229)
is_adjacent × bim_idx = 23	-0.319	-0.447*
	(0.229)	(0.235)
is_adjacent × bim_idx = 24	-0.164	-0.080
	(0.312)	(0.260)
Fixed-effects		
grid_id	Yes	Yes
bim_idx	Yes	Yes
Fit statistics		
Observations	12,160	12,120
Squared Correlation	0.54818	0.58713
Pseudo R^2	0.33976	0.33165
BIC	29,342.9	26,331.0

Clustered (grid_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Treatment = adjacent to Fase1 grid. Post = bim_idx $\geq$ 8. Grid and bimestre FE.
Notes: Poisson QMLE. Event study comparing adjacent vs. non-adjacent never-treated grids. Grid and bimester FE. Cluster SE by grid. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

D.7 Pre-Treatment Balance

Figure 24 plots mean robbery counts per grid cell for Fase 1 (treated) and Fase 2 (not-yet-treated) grids across the six pre-treatment bimesters. Trends are visually parallel and a joint F-test on cohort $\times$ time interactions fails to reject the null of equal pre-trends, supporting the parallel trends assumption underlying the Sun-Abraham estimator.

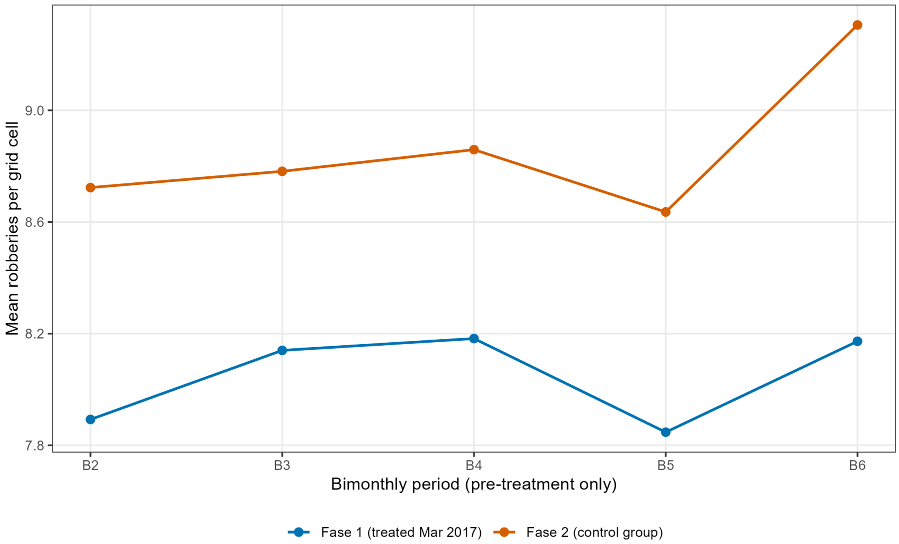

Figure 24: Pre-treatment parallel trends: mean robberies per grid cell, Fase 1 (treated) vs. Fase 2 (not-yet-treated), bimesters 2–6 (Mar 2016–Dec 2016).

D.8 Alternative Outcomes for the event study

Table 18 replicates the baseline Sun-Abraham intensity decomposition replacing *n_robos* with *n_hurtos* and *n_crimes*. Results for *n_crimes* are qualitatively consistent with the baseline; *n_hurtos* yields noisier estimates, consistent with the balance test evidence that hurto levels differ between cohorts in levels but not trends.

Table 18: SA ATT: alternative outcomes (intensity= 1 panel)

Model:	Robos (1)	Hurtos (2)	All crimes (3)
Variables			
Constant	0.108*	-0.403***	0.764***
	(0.058)	(0.103)	(0.065)
bim_idx = -6	-0.012	-0.152	-0.021
	(0.071)	(0.126)	(0.074)
bim_idx = -5	0.054	-0.189	-0.008
	(0.066)	(0.116)	(0.071)
bim_idx = -4	0.069	-0.099	0.027
	(0.068)	(0.117)	(0.072)
bim_idx = -3	0.0010	-0.169	-0.034
	(0.068)	(0.119)	(0.072)
bim_idx = -2	0.0001	-0.169	-0.023
	(0.068)	(0.119)	(0.072)
bim_idx = 0	-0.102	-0.302***	-0.147**
	(0.064)	(0.114)	(0.069)
bim_idx = 1	-0.073	-0.403***	-0.156**
	(0.064)	(0.113)	(0.068)
bim_idx = 2	-0.134**	-0.408***	-0.194***
	(0.067)	(0.120)	(0.071)
bim_idx = 3	-0.080	-0.196*	-0.089
	(0.066)	(0.113)	(0.069)
bim_idx = 4	-0.098	-0.190*	-0.098
	(0.067)	(0.112)	(0.069)
bim_idx = 6	-0.025	-0.287**	-0.111
	(0.064)	(0.116)	(0.070)
bim_idx = 7	0.021	-0.341***	-0.130*
	(0.068)	(0.117)	(0.072)
bim_idx = 8	0.036	-0.255**	-0.134*
	(0.067)	(0.115)	(0.070)
bim_idx = 9	0.050	-0.269**	-0.122*
	(0.064)	(0.113)	(0.069)
bim_idx = 10	0.031	-0.329***	-0.119*
	(0.066)	(0.116)	(0.072)
bim_idx = 12	-0.092	-0.046	-0.061
	(0.062)	(0.114)	(0.068)
bim_idx = 13	-0.064	-0.038	-0.075
	(0.065)	(0.113)	(0.070)
bim_idx = 14	-0.182***	0.010	-0.110
	(0.066)	(0.113)	(0.071)
bim_idx = 15	-0.145**	0.022	-0.084
	(0.062)	(0.112)	(0.069)
bim_idx = 16	-0.759***	-0.580***	-0.584***
	(0.069)	(0.113)	(0.070)
n_bus_stops	0.049***	0.063***	0.052***
	(0.003)	(0.006)	(0.004)
Fit statistics			
Observations	20,520	20,520	20,520
Squared Correlation	0.15760	0.13926	0.20398
Pseudo R^2	0.08849	0.12647	0.13602
BIC	77,523.1	67,974.0	108,277.9

Clustered (grid_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

D.9 Night-Hours Placebo Test

As a falsification exercise, I replicate the baseline Sun-Abraham event study replacing the daytime sample (hours 6–19 h) with night hours (20–23 h and 0–5 h), during which *Agentes de Prevención* have no operational presence and student flows are absent. Under the identifying assumption that the estimated crime reduction operates through route-level guardianship during school transition windows, the programme should exert no effect on night-hour robberies. Panel construction, cohort assignment, and estimator are identical to the main specification.

Pre-treatment coefficients are jointly insignificant ($F = 0.576$, $p = 0.631$), and post-treatment coefficients are equally indistinguishable from zero ($F = 0.573$, $p = 0.778$). The aggregate ATT is -0.003 ($p = 0.952$). This pattern rules out confounding by unobserved shocks correlated with Sendero route placement and is consistent with the guardianship channel as the operative mechanism.

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Overview of AI Tools Used

This thesis was completed with the assistance of the following AI tools:

- **Claude Sonnet 4.6 and Claude Opus 4.7** (Anthropic) — used for code generation, debugging, and formatting in R and LaTeX.
- **Gemini** (Google DeepMind) — used for structural editing and revision of academic text.
- **ChatGPT** (OpenAI) — used for structural editing and revision of academic text.

All AI-generated outputs were critically reviewed, verified, and adapted by the author. The intellectual content, analytical decisions, and conclusions of this thesis remain solely the responsibility of the author.

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Hiermit versichere ich, dass ich die vorliegende Arbeit selbständig und ohne Benutzung anderer als der angegebenen Hilfsmittel angefertigt, noch nicht einer anderen Prüfungsbehörde vorgelegt und noch nicht veröffentlicht habe.

Im Falle der Nutzung von generativen Modellen zur Erstellung von Texten, Abbildungen, Berechnungen und anderen Leistungen verantworte ich die Auswahl, Übernahme und sämtliche Ergebnisse des von mir verwendeten generierten Outputs vollumfänglich selbst. Im Verzeichnis Übersicht verwendeter Hilfsmittel” habe ich alle verwendeten generativen Modelle mit ihrem Produktnamen benannt und angegeben, wie, in welchem Umfang und zu welchem Zweck diese benutzt wurden.

I hereby declare that I have prepared this thesis independently and without the use of aids other than those specified, that I have not yet submitted it to another examination authority and that it has not yet been published.

In the case of the use of generative models for the creation of texts, illustrations, calculations and other services, I am fully responsible for the selection, adoption and all results of the generated output used by me. In the list “Overview of tools used” I have named all generative models used with their product name and indicated how, to what extent and for what purpose they were used.

25.04.26

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